Innovation partner H2S HACKERSKILL

Media partner YOURSTORY

AI for Bharat Hackathon

Powered by aws

Team Name : eGramSabha

Team Leader Name : Anshul Jain

Problem Statement : PS03 - AI for Rural Innovation & Sustainable Systems

eGramSabha

"Sashakt Panchayat, Samriddh Bharat"

AI-Powered Digital Gram Sabha Platform

GitHub: https://github.com/anshuljain90/ai4bharat-empowerpanchayat
Demo Video: https://drive.google.com/drive/folders/1aLeLtSPlDUaoypi3TAu2tg4I-TpxVUNH
Live Demo: https://44.194.253.143/
Admin Login: `admin / AdminPassword123`
Official Login: `password same as configured username`

AWS AI for Bharat Hackathon

Problem Statement

In 1992, India passed the 73rd Constitutional Amendment — a radical act of faith in its own people. It created 250,000+ Gram Panchayats and mandated Gram Sabhas as the foundation of grassroots democracy. Three decades later, that promise remains largely unfulfilled.

5-10%
Average Gram Sabha attendance

<3%
Women participation in many states

60%+
Rural Indians with limited literacy

30%
Eligible citizens excluded from welfare schemes

ZERO
Digital trail of Gram Sabha decisions

Barrier	Impact
Paper agendas on notice boards	Low awareness — excludes remote hamlets, women at home, daily wage workers
Paper sign-in sheets	Easy to forge — routinely inflated to show higher attendance numbers
Hand-written minutes by clerk	Subjective, often incomplete, sometimes pre-written before the meeting
Single language proceedings	Excludes tribal & minority communities — Gondi, Santhali, Bhili speakers silenced
No issue tracking system	Promises forgotten between meetings — citizens stop attending
Physical presence required	Excludes women (social norms), elderly (mobility), differently-abled citizens

Solution Overview

eGramSabha is an AI-driven, voice-first, face-authenticated platform that digitizes the entire Gram Sabha lifecycle — from citizen registration to issue resolution tracking. Three role-based portals (Admin, Official, Citizen) serve every stakeholder.

Admin Portal
Panchayat CRUD, CSV import, official management (6 roles), letterhead, platform config

Official Portal
Citizen registration, issue management, meeting scheduling, AI agenda, attendance, PDF exports

Citizen Portal
Face login, voice/text issue reporting, RSVP, issue tracking, support tickets

5-Step Lifecycle

1. REGISTER Face-authenticated citizen onboarding → Citizens enrolled →
2. REPORT Voice-first issue submission → Issues collected →
3. MEET AI-generated agenda & scheduled Sabha → Sabha conducted →
4. RECORD AI Minutes of Meeting & attendance → Decisions captured →
5. TRACK Issue lifecycle & status updates

5-Step Gram Sabha Lifecycle

1. REGISTER
Face

Admin adds citizens via CSV or manual entry. Officials onboard via face biometric. 128-dim face descriptor stored — no passwords, no OTPs.

2. REPORT
Voice

Citizens record voice complaints in their native language, anytime. AI transcribes, corrects grammar, translates, analyses sentiment, assigns priority.

3. MEET
Agenda

Officials schedule Gram Sabha. AI summarises issues into backlog. Officials select items for agenda. Translated to multiple languages. TTS read-aloud.

4. RECORD
Attendance

Officials take facial biometric attendance to meet quorum threshold. Meeting auto-transitions to IN_PROGRESS when quorum met.

5. TRACK
Status

Every issue tracked: REPORTED → PICKED_IN_AGENDA → DISCUSSED_IN_GRAM_SABHA → RESOLVED. Officials add resolution remarks. Zero promises lost.

Citizen Portal

Face Login

Last 4 digits of voter ID + face scan. Euclidean distance < 0.4. No passwords needed.

Voice Issue Reporting

Record complaint in native language. 6 categories, 20+ subcategories. AI transcribes and translates.

Issue Tracking

Real-time status: REPORTED → PICKED_IN_AGENDA → DISCUSSED → RESOLVED.

Meeting RSVP

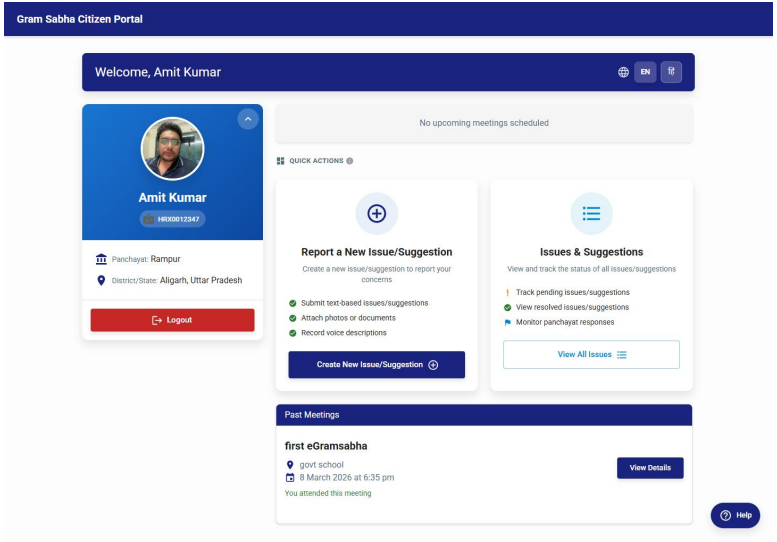
View upcoming Gram Sabhas. RSVP to confirm attendance.

Multilingual UI

English and Hindi interface. AI content in 10+ languages.

Support Tickets

Raise help desk tickets. Anonymous submission for vulnerable reporters.



Official Portal

Citizen Registration

Register citizens with voter ID + face capture. Liveness: 2 blinks + 5 head movements.

Issue Management

Filter by status, category, priority, ward. View transcriptions, sentiment, summaries.

Meeting Scheduling

Set date, time, location, expected duration. SMS notifications to citizens.

AI Agenda Generation

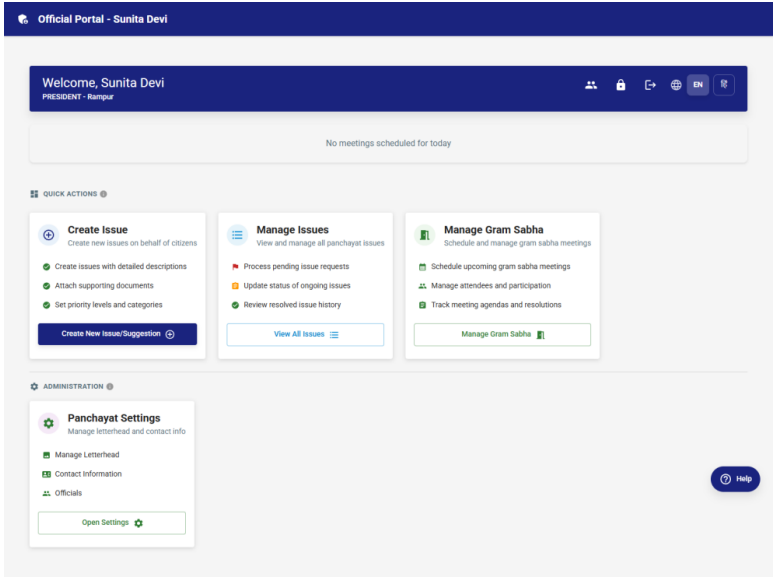
View AI-summarised issue backlog. Select items for agenda. Translate and TTS read-aloud.

Biometric Attendance

Mark citizen attendance via face verification or manual check-in. Quorum auto-check.

PDF Exports

Attendance reports, agenda documents with custom panchayat letterhead.



Admin Portal

Panchayat Management

Create/edit panchayats with LGD code. Ward management. Letterhead upload and preview.

Official Management

Register officials with 6 roles: Secretary, President, Ward Member, Committee Secretary, Guest, Admin.

Bulk CSV Import

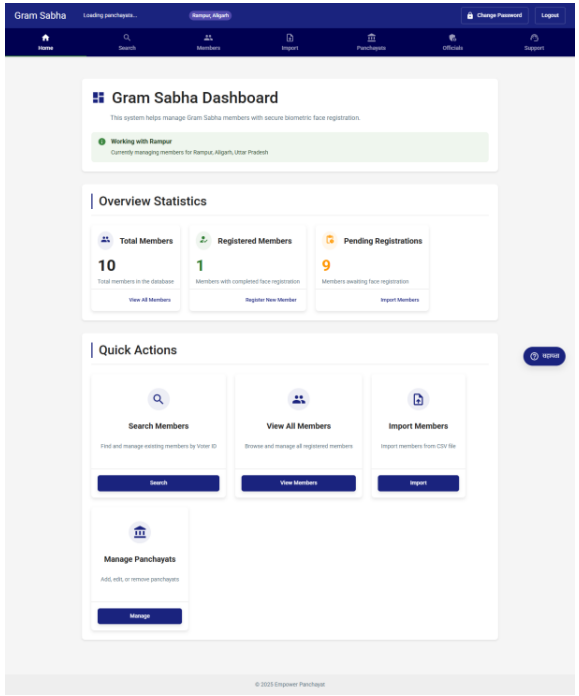
Import citizens, officials, and panchayats via CSV upload. Batch operations.

Platform Configuration

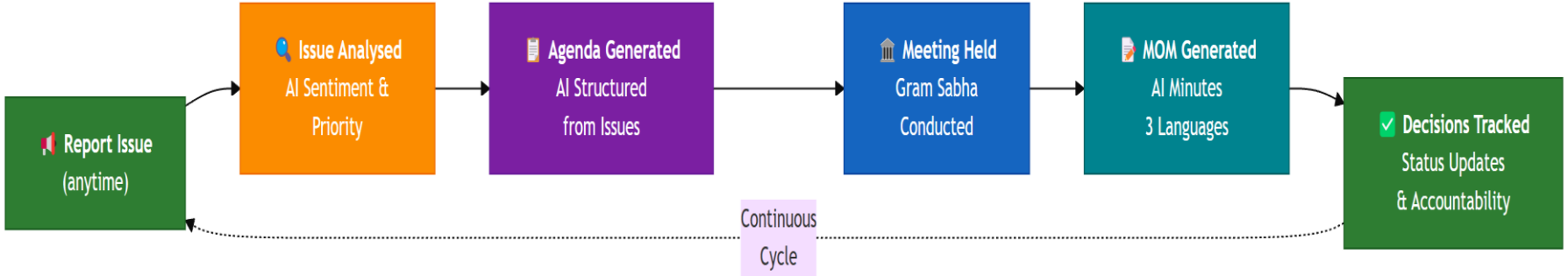
Sabha quorum criteria, feature toggles, system-wide settings.

Support Ticket Management

View and respond to citizen/official help desk tickets.



Continuous Governance Loop



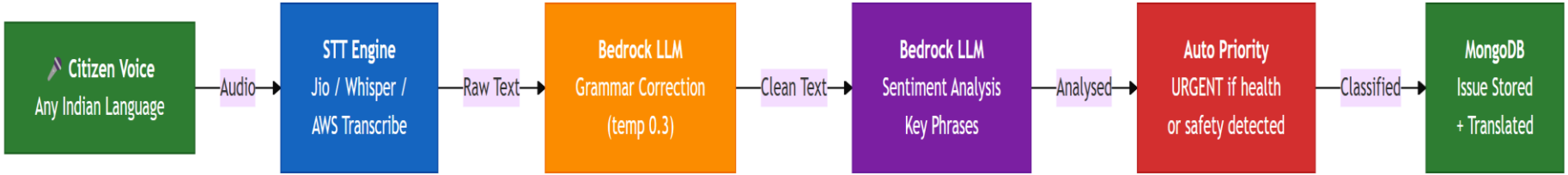
Report	Citizens report issues via voice in any language, any time — not tied to meeting dates.
Summarise	AI categorises, deduplicates, and summarises issues hourly via cron jobs.
Agenda	Officials review AI summaries and select items for the next Gram Sabha agenda.
Discuss	Face-verified attendance, quorum check, structured discussion of agenda items.
Resolve	Resolution remarks recorded. Undiscussed items return to backlog automatically.

Citizens can report issues at any time — the cycle is continuous, not tied to meeting dates

Why AI? — Before vs After

User	Without AI	With eGramSabha AI
Illiterate citizen (60%+ of rural India)	Cannot report issues — requires writing in complaint register during office hours	Speaks complaint in any language. AI transcribes, corrects, translates, assigns priority.
Panchayat secretary	Hand-writes minutes for hours, often days later. Subjective, sometimes pre-written.	AI generates structured MOM (Planned): Discussion Points, Decisions, Action Items. 3 languages.
Women at home (~3% participation)	Must attend male-dominated meeting. Social norms discourage speaking.	Reports issues via voice from home. Same formal weight. Face auth — no password.
Official creating agenda	Ad-hoc list influenced by powerful voices. No systematic inclusion.	AI structures backlog from ALL citizen issues. Officials select. Equal weight.
Tribal/minority citizen	Excluded by state-language proceedings. Cannot understand.	Speaks native language. AI translates. Agenda in their language. TTS reads aloud.
Block/District admin	Physical inspection needed. Paper records inaccessible.	Digital records, attendance data, issue analytics. Audit remotely.

Voice Issue Reporting Pipeline



1. **Citizen records voice:** Native language, 6 categories, 20+ subcategories
2. **Audio uploaded:** Stored in S3 or GridFS as issue attachment
3. **STT transcription:** Provider-agnostic: Jio / Whisper / AWS Transcribe
4. **Grammar correction:** Bedrock Nova Lite corrects and enhances
5. **3-language output:** enhanced_original + enhanced_english + enhanced_hindi
6. **Sentiment analysis:** Bedrock: sentiment label, scores, key phrases (top 10)
7. **Auto-priority:** Health/safety + strong negative → URGENT auto-upgrade
8. **Results stored:** MongoDB: issue.sentiment, issue.keyPhrases, issue.priority

AI Issue Summarisation & Agenda Generation

AI Summarisation (Hourly Cron)

1. Cron job runs hourly (summaryCronJobs.js) — checks for unsummarised issues in MongoDB
2. Calls video-mom-backend /summarize-issues endpoint with batch of new issues
3. Bedrock Nova Lite categorises by type, deduplicates similar issues, generates concise summaries
4. Sentiment + priority carried forward; URGENT issues flagged for immediate attention
5. IssueSummary records created in MongoDB — linked to original issues for traceability

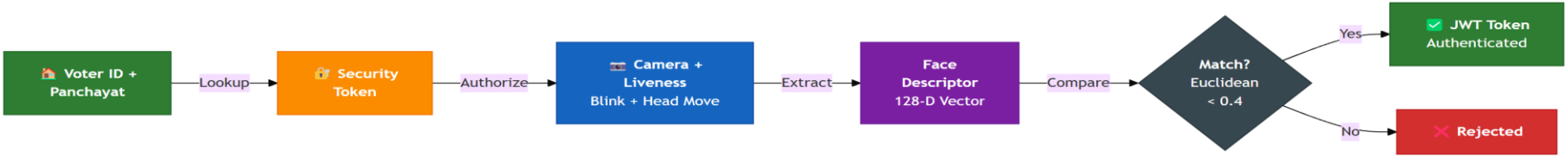
Agenda Generation (Official-Driven)

1. Officials view IssueSummary backlog — filterable by category, ward, priority, date
2. Officials select items for agenda; AI structures numbered items with citizen context
3. AWS Translate converts agenda to Hindi + regional languages (auto-detect source)
4. Amazon Polly synthesises TTS audio; S3 cache: tts-cache/{lang}/{sha256}.mp3
5. Citizens can Read Aloud agenda in their language before the meeting
6. Post-meeting: undiscussed items return to backlog automatically for next Sabha

Source: summaryCronJobs.js → /api/summarize-issues → Bedrock → IssueSummary → Agenda → Translate → Polly TTS

AI summarises — Officials decide. Every citizen voice gets equal representation.

Face Authentication Pipeline



Registration Flow

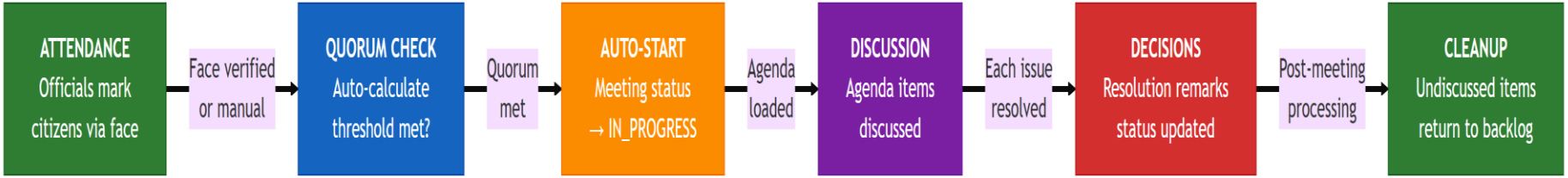
1. Official/Admin registers citizen with voter ID
2. Liveness: 2 blinks + 5 head movements (MediaPipe FaceMesh)
3. face-api.js extracts 128-dim face descriptor client-side
4. Face image → S3, thumbnail generated via Sharp
5. Descriptor stored in MongoDB user record

Login Flow

1. Enter last 4 digits of voter ID
2. Camera capture (no liveness on login)
3. New face descriptor computed via face-api.js
4. Euclidean distance: threshold < 0.4
5. Security token: random → SHA256 hash (anti-replay)
6. On match: JWT issued (citizen 72h, refresh 7d)

Production: Rekognition: IndexFaces per panchayat, SearchFacesByImage at 90% similarity. Non-blocking fallback to local.

Meeting Day Flow



1. Officials Take Attendance: Face-verified or manual check-in. Verification method recorded.

2. Quorum Check: attendanceCount >= ceil(totalVoters × sabhaCriteria / 100)

3. Meeting Auto-Starts: Status: SCHEDULED → IN_PROGRESS when quorum met.

4. Issues Discussed: Agenda items discussed. Status: PICKED_IN_AGENDA → DISCUSSED.

5. Decisions Recorded: Resolution remarks added. RESOLVED / TRANSFERRED / NO_ACTION_NEEDED.

6. Post-Meeting: Undiscussed items return to backlog for next meeting.

MOM Generation (Planned): meeting audio → STT → Bedrock structured MOM → 3 languages

TTS & Multilingual Support

TTS Caching Flow (Amazon Polly)

1. Agenda text ready (English / Hindi / Regional)
2. Generate SHA256 hash of text content
3. Check S3 cache: tts-cache/{language}/{hash}.mp3
4. Cache hit → return cached audio instantly
5. Cache miss → Polly synthesises (Aditi Hindi, Raveena English)
6. Store in S3, return audio. Same text never re-synthesised.

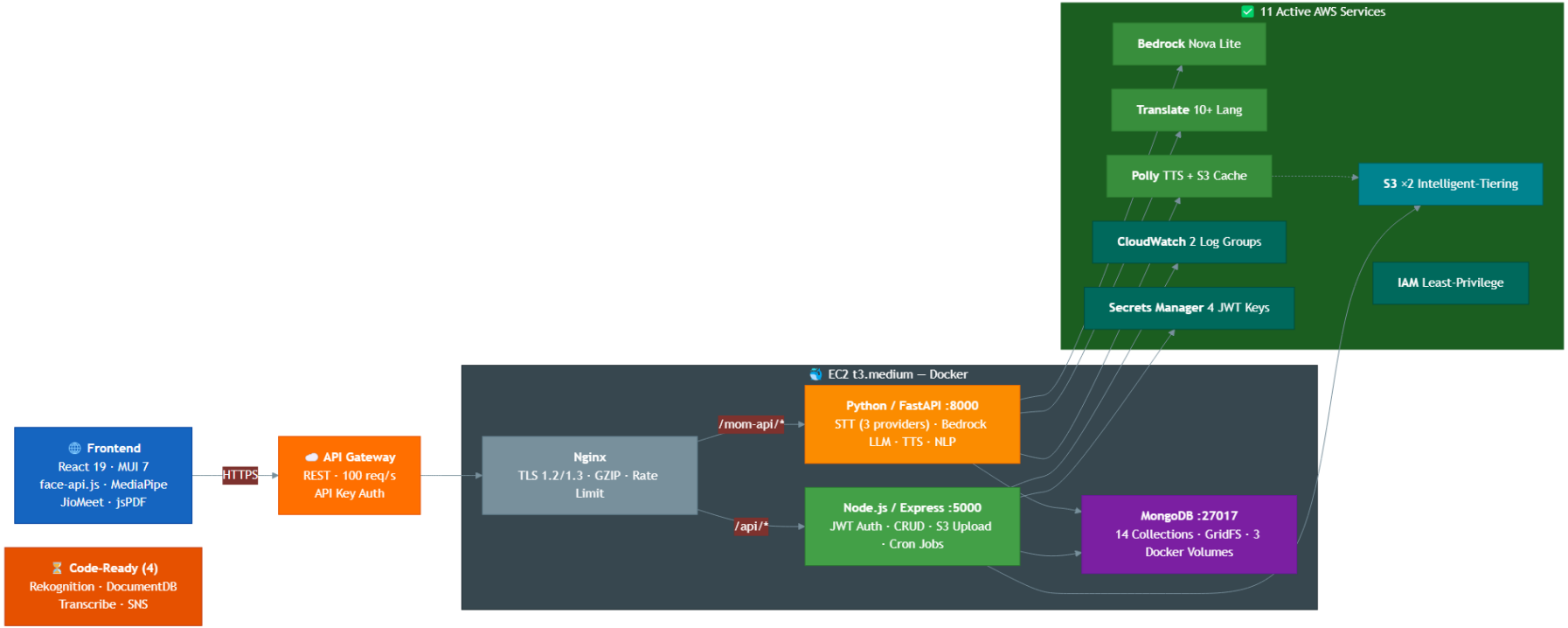
Supported Languages (10+)

Hindi	Tamil
Telugu	Kannada
Malayalam	Bengali
Gujarati	Marathi
Punjabi	English

UI: English and Hindi UI via React Context with static translation files

AI Content: AI content (agendas, summaries, transcriptions) translated via AWS Translate across 10+ languages

System Architecture



4 containers on bridge network: frontend, backend, video-mom-backend, mongodb

11 Active AWS Services

4 Code-Ready AWS Services

Architecture Layers

Layer	Components & Details
Browser (Client)	React 19, MUI 7, face-api.js, MediaPipe FaceMesh, JioMeet SDK, jsPDF
API Gateway	eGramSabha-MOM-API, REST REGIONAL, 100 req/s, API key, 100K/day quota
Nginx Reverse Proxy	TLS 1.2/1.3, GZIP, 10 req/s per IP, security headers, 500MB upload
Node.js / Express :5000	JWT auth (4 secrets), CRUD, S3 upload, SNS, cron jobs, Winston logging
Python / FastAPI :8000	STT (3 providers), Bedrock LLM, Polly TTS, Translate, sentiment analysis
MongoDB :27017	14 collections, GridFS, DocumentDB-compatible, 3 Docker volumes

Security Architecture — 6 Layers

1. API Gateway	API key (x-api-key) + 100 req/s rate limit + 200 burst + 100K/day quota
2. Nginx	TLS 1.2/1.3 + 10 req/s per IP + X-Frame-Options + HSTS + XSS-Protection
3. Express Middleware	Helmet + CORS + xss-clean + hpp + content security
4. JWT Authentication	3 role-specific secrets: Admin 12h, Official 24h, Citizen 72h, Refresh 7d
5. RBAC Authorization	6 roles with per-resource permissions, ward-scoped data access
6. Secrets Manager	Zero hardcoded credentials. egramsabha/prod secret with 4 JWT keys.

RBAC Roles

ADMIN | SECRETARY | PRESIDENT | WARD_MEMBER | COMMITTEE_SECRETARY | GUEST | CITIZEN

Traffic Flow

CRUD Flow

Browser → API Gateway → Nginx :443 → /api/* → Node.js :5000

Auth, issue CRUD, file uploads, citizen management

AI Flow

Browser → API Gateway → Nginx :443 → /mom-api/* → FastAPI :8000

TTS synthesis, agenda generation, sentiment analysis

Cron Flow

Node.js :5000 → Docker internal → FastAPI :8000

Hourly summarisation, transcription retry, STT + grammar correction

Docker Volumes

mongodb_data
Persistent database storage

backend_uploads
Face photos, attachments, letterheads

video_mom_temp_storage
Temporary audio for AI processing

How AWS — 15 Services

Service	Configuration	Usage
Amazon Bedrock	Nova Lite, Converse API	Grammar correction, summarisation, sentiment, translation fallback
Amazon S3	2 buckets, Intelligent-Tiering	Face photos, TTS cache, letterheads, attachments
AWS Translate	Auto-detect source	AI content across 10+ Indian languages
Amazon Polly	Aditi + Raveena	TTS read-aloud with SHA256 S3 caching
API Gateway	REST, REGIONAL	Single entry point, 100 req/s, API key auth
CloudWatch	2 log groups	Winston + watchtower, JSON logs, 30-day retention
Secrets Manager	egramsabha/prod	4 JWT secrets, graceful fallback to env vars
EC2	t3.medium	Docker Compose host, burstable CPU for AI spikes
EBS	20 GB gp3	3 persistent Docker volumes
Elastic IP	Static address	DNS and API Gateway endpoint
IAM	Least-privilege	Per-service access policies

Code-Ready (4 Services)

Rekognition
Server-side face verification, 90% threshold, per-panchayat collections

AWS Transcribe
24+ languages, async batch via S3

DocumentDB
Managed MongoDB, multi-AZ, automated backups

SNS
SMS notifications, requires TRAI DLT registration

Provider Abstraction — Zero Lock-In

Environment Variable	Options	Purpose
STT_PROVIDER	jio / whisper / aws_transcribe	Speech-to-Text engine
LLM_PROVIDER	bedrock / huggingface	Large Language Model backend
TRANSLATION_PROVIDER	aws_translate / llm	Translation engine
TTS_PROVIDER	polly / disabled	Text-to-Speech service
STORAGE_BACKEND	s3 / gridfs	File storage backend
FACE_VERIFICATION_PROVIDER	local / rekognition	Face matching engine
USE_DOCUMENTDB	true / false	MongoDB → DocumentDB switch
COMPREHEND_ENABLED	true / false	AWS Comprehend for NLP

Swap any service without code changes. Move from hackathon to production by flipping environment variables.

Technologies Utilized

Frontend

- **React 19 + MUI 7:** SPA with Material UI components
- **face-api.js 0.22.2:** Client-side face recognition (128-dim)
- **MediaPipe FaceMesh:** Liveness detection (blink + head)
- **JioMeet SDK:** Video meeting integration
- **jsPDF + html2pdf.js:** PDF generation with Hindi fonts

Backend API

- **Node.js 20 + Express 4:** RESTful API, JWT auth, cron jobs
- **Mongoose 8:** MongoDB ODM, 14 collection schemas
- **Sharp:** Image processing, thumbnails
- **Winston:** Structured logging → CloudWatch

AI Backend

- **Python 3.11 + FastAPI:** Async AI pipelines
- **Motor:** Async MongoDB driver
- **boto3:** AWS SDK: Bedrock, S3, Polly, Translate
- **ffmpeg-python:** Audio format conversion

Database

- **MongoDB 7:** 14 collections, GridFS
- **DocumentDB-compatible:** Schemas ready for migration
- **3 Docker volumes:** Persistent data, uploads, temp

Cost — $15–20/month

Service	Monthly Cost	Notes
EC2 (t3.medium)	$12–30	Burstable CPU for AI spikes
EBS (20 GB gp3)	$1.60	3 Docker volumes
Elastic IP	$0	Free while attached
S3 (2 buckets)	$0.50	Intelligent-Tiering
Bedrock (Nova Lite)	$2–5	$0.06/1K input, $0.24/1K output
Translate	$0–1	2M chars/mo free tier
Polly	$0–0.50	5M chars/mo free, S3 cache
CloudWatch	$0–0.50	5 GB/mo free tier
Secrets Manager	$0.40	1 secret
API Gateway	$0	1M calls/mo free tier
TOTAL	$15–20/month	Hackathon scale (1-5 panchayats)

Cost Optimizations

S3 Intelligent-Tiering	Auto-moves cold face images and audio to cheaper storage tiers
Polly S3 Caching	SHA256 content hash as key — same text never re-synthesised
API Gateway Throttling	100 req/s protects expensive Bedrock AI endpoints from abuse
Bedrock Low Temp (0.3)	Shorter, focused responses = fewer output tokens = lower cost
Chunked Processing	1500-char chunks with 200 overlap — handles hour-long transcripts
Transcribe Cleanup	Auto-delete temp S3 objects and jobs after completion

Scale Projections

Phase	Scale	Changes Required	Monthly Cost	Per Citizen/Yr
Hackathon	1–5 panchayats	Current config	$15–20	—
Pilot	100 panchayats	STORAGE_BACKEND=s3, USE_DOCUMENTDB=true	$320	~$3/citizen/yr
State	5K panchayats	ECS Fargate, CloudFront, multi-AZ DocumentDB	$3,250	~$0.80/citizen/yr
National	250K panchayats	Multi-region, service mesh, Bedrock provisioned	$30–37K	$0.60–0.72/citizen/yr

~100x Cost Reduction

Physical Gram Sabha: Rs 500+/meeting vs Digital eGramSabha: Rs 5/citizen/year

At national scale: Rs 0.60 per citizen per year — less than printing one notice board poster.

Impact Potential — What Digital Gram Sabha Can Achieve

Metric	National Average	With Digital Gram Sabha	Improvement
Voter registration	~10% nationally	81% (650 of 801 eligible)	8x
Issues submitted	Handful (verbal, unrecorded)	100+ issues in 3 days	10x+
Women participation	Below 3% in many states	Approximately 50%	16x
Decisions recorded	Few, unrecorded or pre-written	Multiple, digitally recorded	Permanent
Time to minutes	Days later (if at all)	AI-generated instantly	Instant

Registration Approach: Door-to-door registration with tablets, female enumerators for women's participation, elder-friendly assisted registration.

Measurable Outcomes — From a Single Digital Gram Sabha

Garbage Management
Structured tenant fee collection system approved by citizens

Road Repairs
Multiple road repairs prioritised and scheduled by citizen consensus

Welfare Allocation
Transgender welfare: Rs 2,100 minimum approved by the sabha

Clean Energy
24-acre solar power plant approved on unused community land

Self-Governance
Citizens independently levied local taxes and contributed resources

Community Mobilisation
Private company donated a tractor based on community initiative

Fraud Prevention
Digital attendance eliminated paper sign-in fraud

Democratic Inclusion
AI-generated agenda ensured every citizen's issue was formally considered

One digitally-enabled Gram Sabha produced more accountable decisions than years of paper-based proceedings.

National Impact Projection

Impact Area	Mechanism	Projected Outcome
Citizen Participation	Voice-first, face auth, phone-based access	5% → 30%+ attendance
Women's Empowerment	Remote voice reporting, no public speaking needed	3% → 50% participation
Welfare Delivery	Citizen-verified beneficiary lists, digital tracking	Reduce 30% exclusion error
Transparency	AI-generated records, searchable & auditable	Every decision verifiable
Accountability	Issue lifecycle: REPORTED → DISCUSSED → RESOLVED	Zero promises lost
Cost Reduction	Digital vs paper-based at scale	~100x cost reduction

If participation increases from 5% to 25–30%: 125–175 million additional citizens actively participating in local governance. Digital records for every meeting in every panchayat — the largest governance transparency dataset in the world.

Multiplier Effects

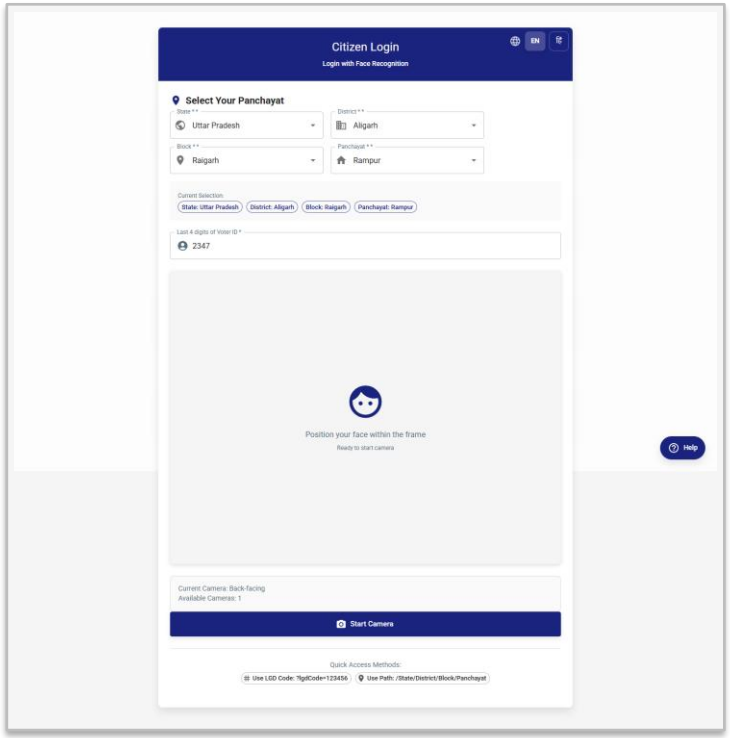
- Reduced rural-urban migration — local issues resolved locally
- Rs 25+ lakh crore welfare spending — active participation reduces leakage
- Better welfare delivery — citizen-verified beneficiary lists
- Youth engagement — transparent, mobile-first governance

Citizen Portal — Face Login

Citizen Portal

Citizen Login

Citizens log in using just the last 4 digits of their voter ID and a face scan. No passwords, no OTPs. The camera captures a live face descriptor and matches it against the registered 128-dim biometric using Euclidean distance < 0.4.



Key Highlights

- Passwordless login — voter ID + face scan only
- Panchayat selection dropdown for multi-panchayat support
- face-api.js client-side matching, no server round-trip for speed

Citizen Portal — Dashboard

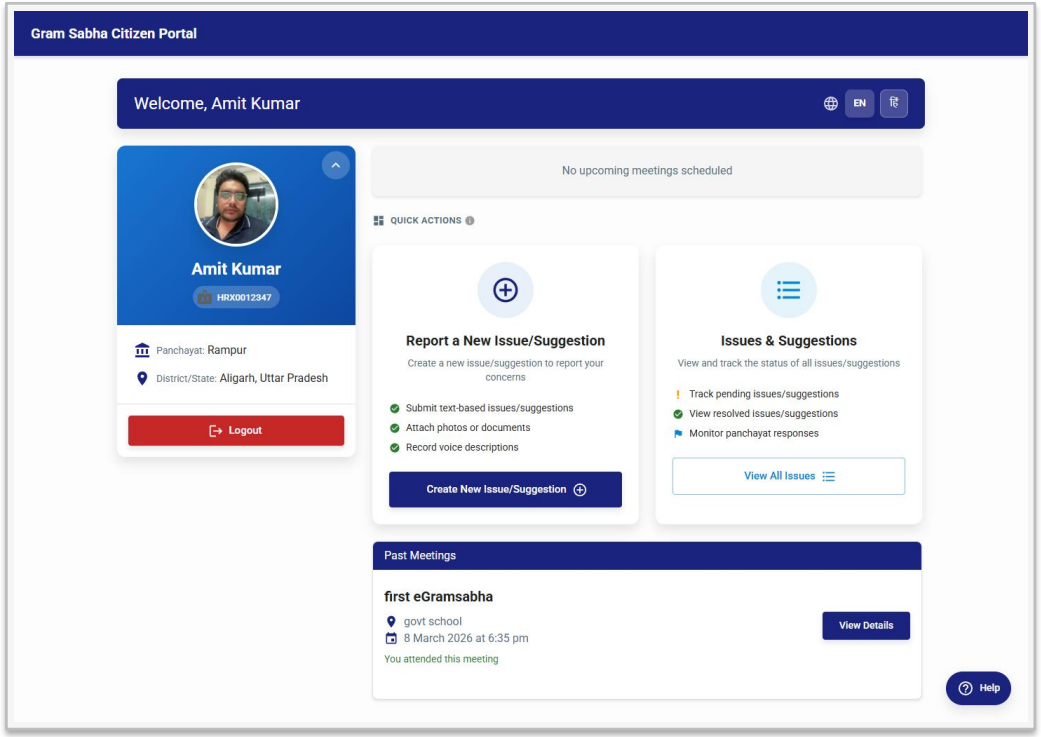
Citizen Portal

Citizen Dashboard

The citizen's home screen after face login. Shows a personalised welcome card, quick action buttons for raising issues and checking meeting schedules, and a summary of past meetings and reported issues.

Key Highlights

- Personalised welcome with citizen name and panchayat
- Quick actions: Report Issue, View Meetings, Track Issues
- Past meeting history and issue status at a glance



Citizen Portal — Report Issue

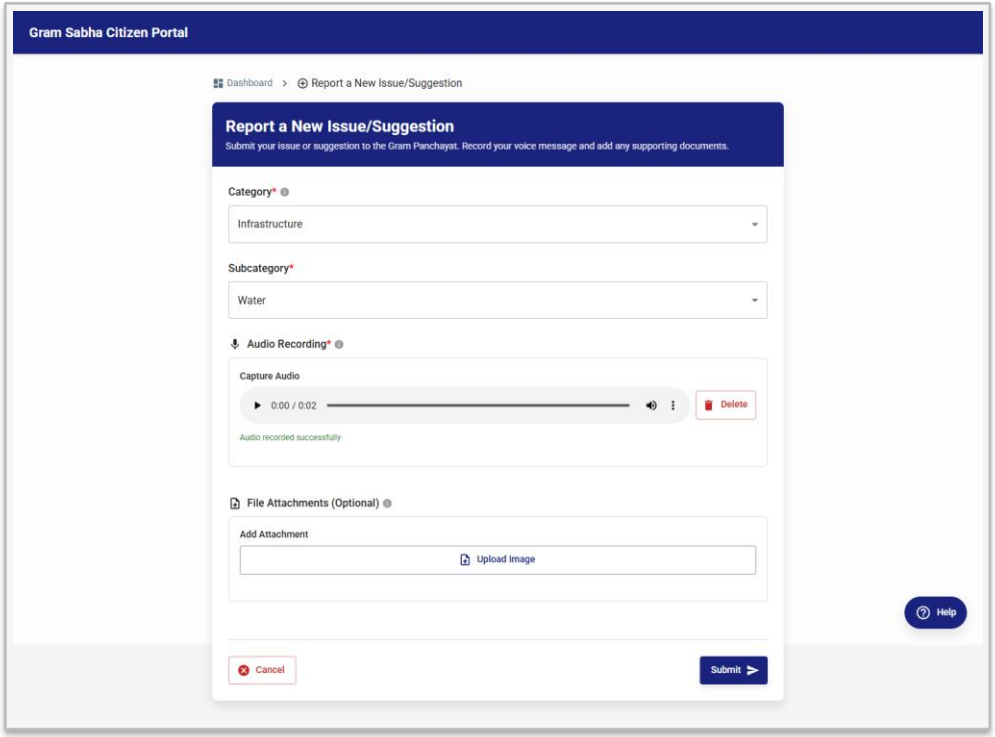
Citizen Portal

Create Issue / Suggestion

Citizens can report issues by recording a voice message in their native language or typing text. Choose from 6 categories and 20+ subcategories. AI transcribes, corrects grammar, translates, and assigns priority automatically.

Key Highlights

- Voice recording in any native language
- 6 categories: Infrastructure, Health, Education, Water, Sanitation, Other
- AI auto-transcription, grammar correction, and translation



Citizen Portal — My Issues

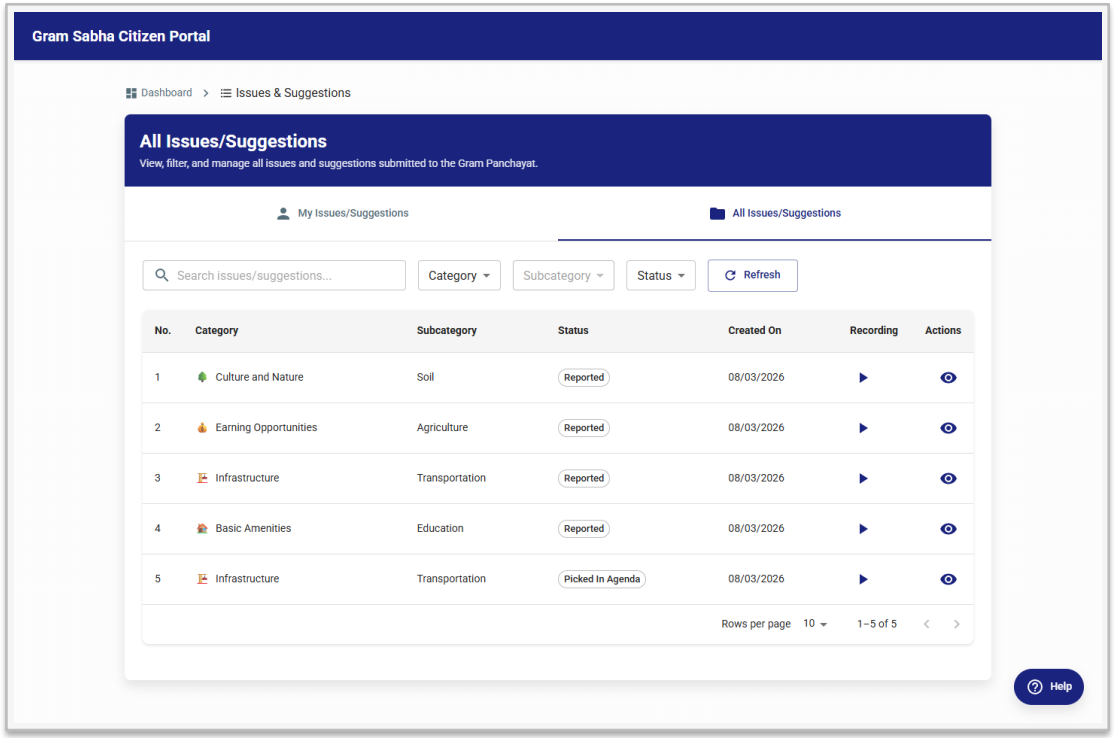
Citizen Portal

All Issues & Suggestions

Citizens can track every issue they have reported with real-time status updates. Each issue shows its current lifecycle stage from REPORTED through to RESOLVED, along with priority level and category.

Key Highlights

- Real-time status: REPORTED → PICKED_IN_AGENDA → DISCUSSED → RESOLVED
- Priority indicators and sentiment badges
- Full transparency — no promises lost between meetings



Official Portal — Login

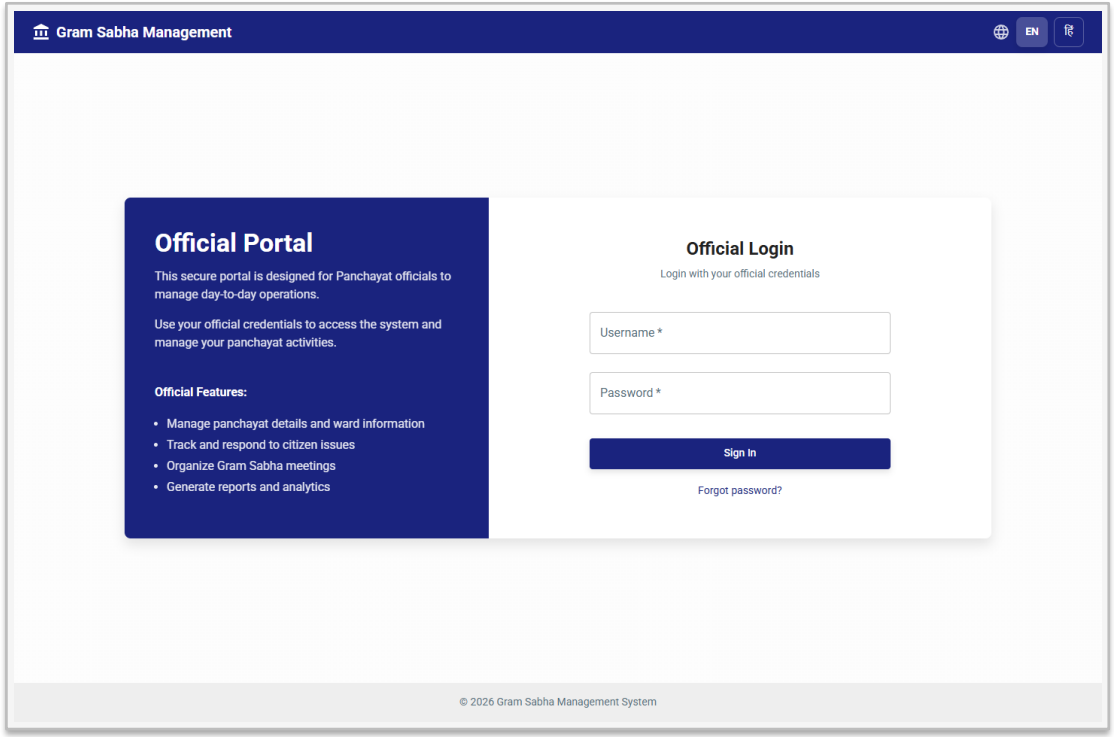
Official Portal

Official Login

The Official Portal login page in English. Officials authenticate using their credentials with role-based JWT tokens. Supports 6 roles: Secretary, President, Ward Member, Committee Secretary, Guest, and Admin.

Key Highlights

- Role-based JWT authentication with per-role token expiry
- Secretary 24h, President 24h, Ward Member 24h token lifetime
- Automatic redirect to role-appropriate dashboard



Official Portal — Login (Hindi)

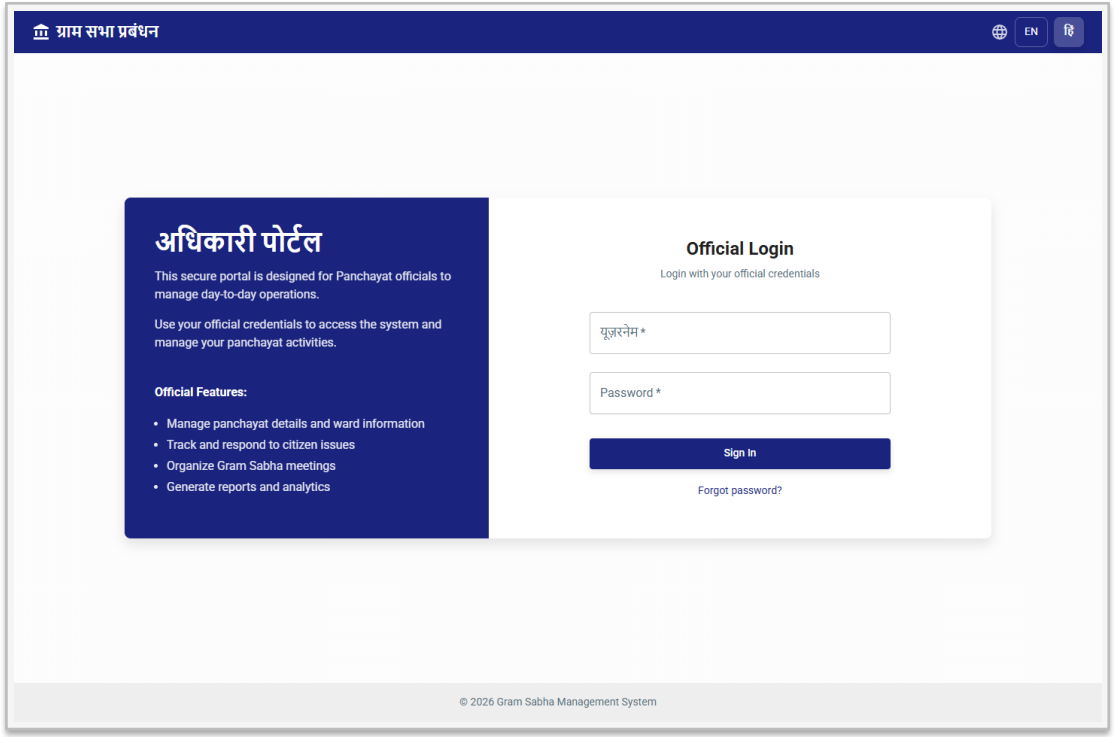
Official Portal

Official Login — Hindi

The same Official Portal login page rendered in Hindi, demonstrating full i18n support. Language toggle is available on every page. All UI labels, placeholders, and validation messages are translated.

Key Highlights

- Complete Hindi translation via React Context + static files
- One-click language toggle persisted in local storage
- All form validation messages also translated



Official Portal — Dashboard

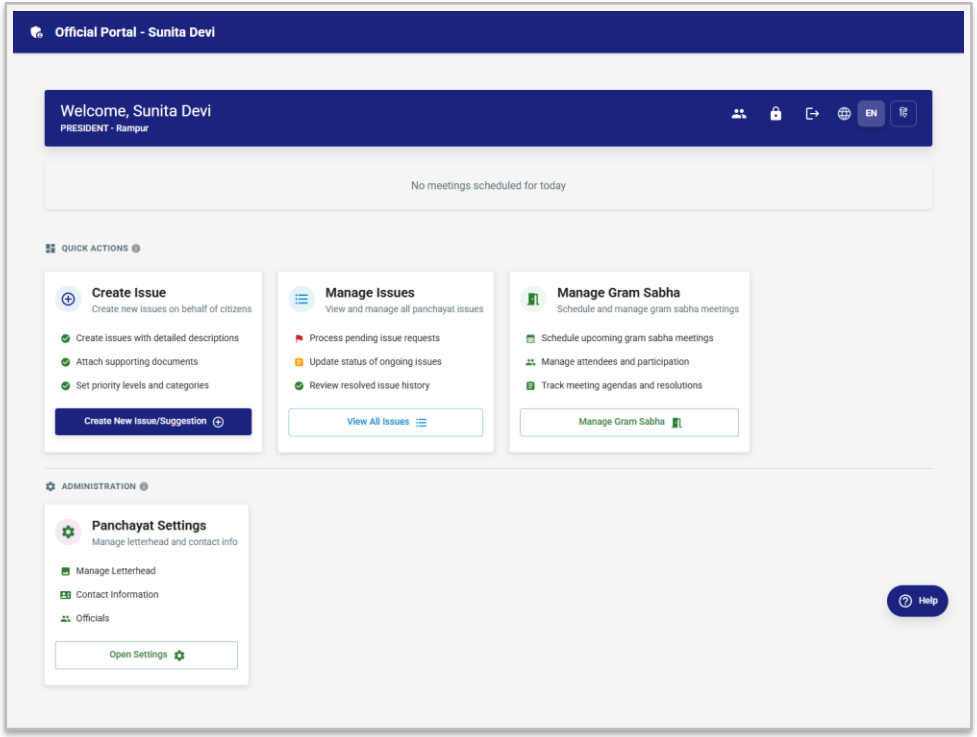
Official Portal

Official Dashboard

The official's command centre showing quick action cards for all key workflows: citizen registration, issue management, Gram Sabha scheduling, and attendance tracking. Provides at-a-glance statistics and pending action items.

Key Highlights

- Quick actions: Register Citizens, Manage Issues, Schedule Meetings
- At-a-glance statistics for pending and resolved items
- Role-based access: Secretary, President, Ward Member, etc.



Official Portal — Dashboard (Hindi)

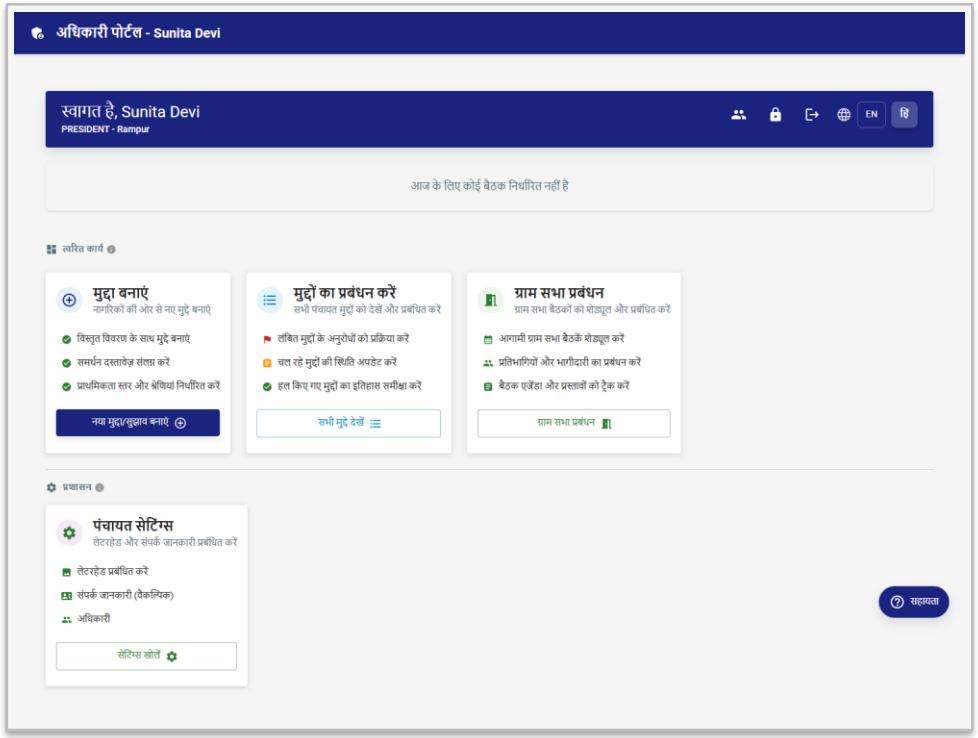
Official Portal

Dashboard — Hindi

The entire Official Portal dashboard rendered in Hindi, demonstrating full i18n support. Every label, button, card title, and status indicator is translated using static translation files via React Context.

Key Highlights

- Complete UI translation: English ↔ Hindi toggle
- AI content translated to 10+ Indian languages via AWS Translate
- TTS read-aloud for agenda items via Amazon Polly



Official Portal — Create Issue

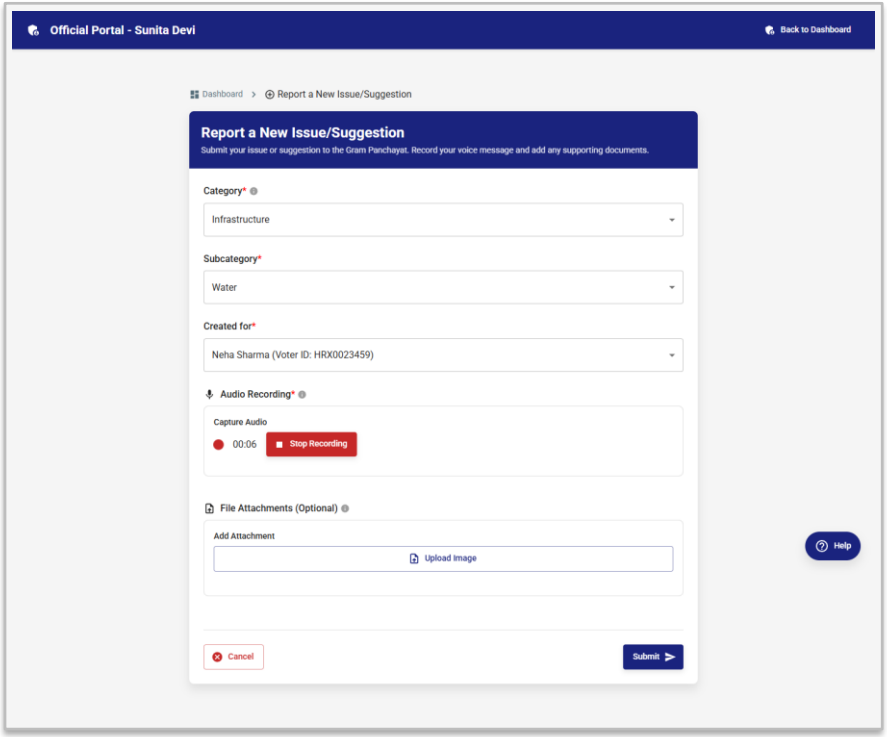
Official Portal

Create Issue / Suggestion

Officials can also create issues on behalf of citizens who visit the panchayat office. The form captures category, subcategory, description, and optional voice recording. Issues are linked to the citizen's profile for tracking.

Key Highlights

- 6 categories with 20+ subcategories for structured reporting
- Voice recording or text input with file attachments
- Auto-assigned priority based on category and sentiment



Official Portal — Create Issue (Hindi)

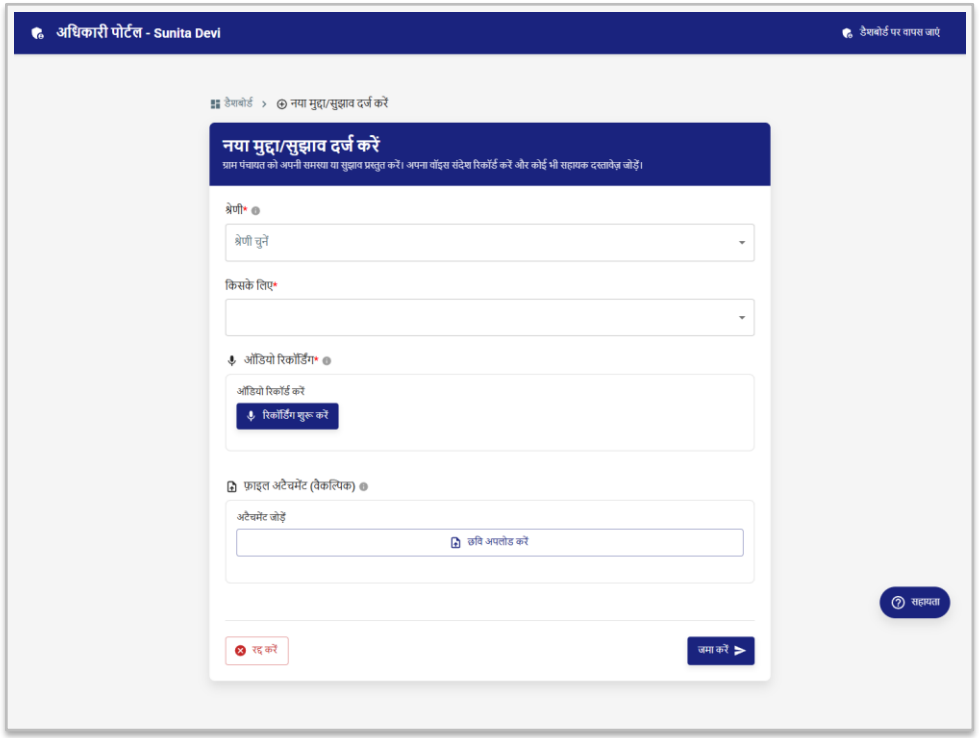
Official Portal

Create Issue — Hindi

The Create Issue form in Hindi, showing complete translation of all form fields, dropdown options, category labels, and action buttons. Officials comfortable in Hindi can operate the entire workflow in their preferred language.

Key Highlights

- All form labels and dropdowns translated to Hindi
- Category and subcategory options in Hindi
- Consistent UI layout across both languages



Official Portal — Issue Management

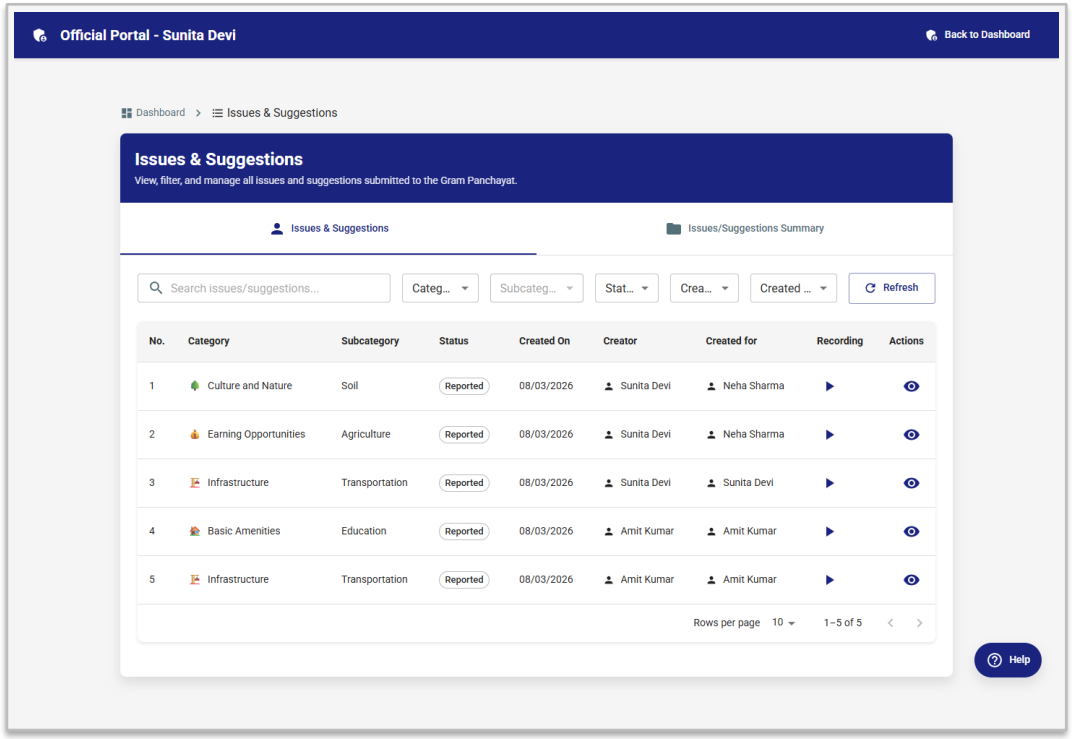
Official Portal

Issues & Suggestions

A filterable table of all citizen-reported issues with status, category, priority, and ward filters. Officials can view AI transcriptions, sentiment analysis results, and manage issue lifecycle from this screen.

Key Highlights

- Filter by status, category, priority, ward, and date range
- AI-generated transcription and sentiment visible per issue
- Bulk operations for agenda selection



Official Portal — AI Issue Summary

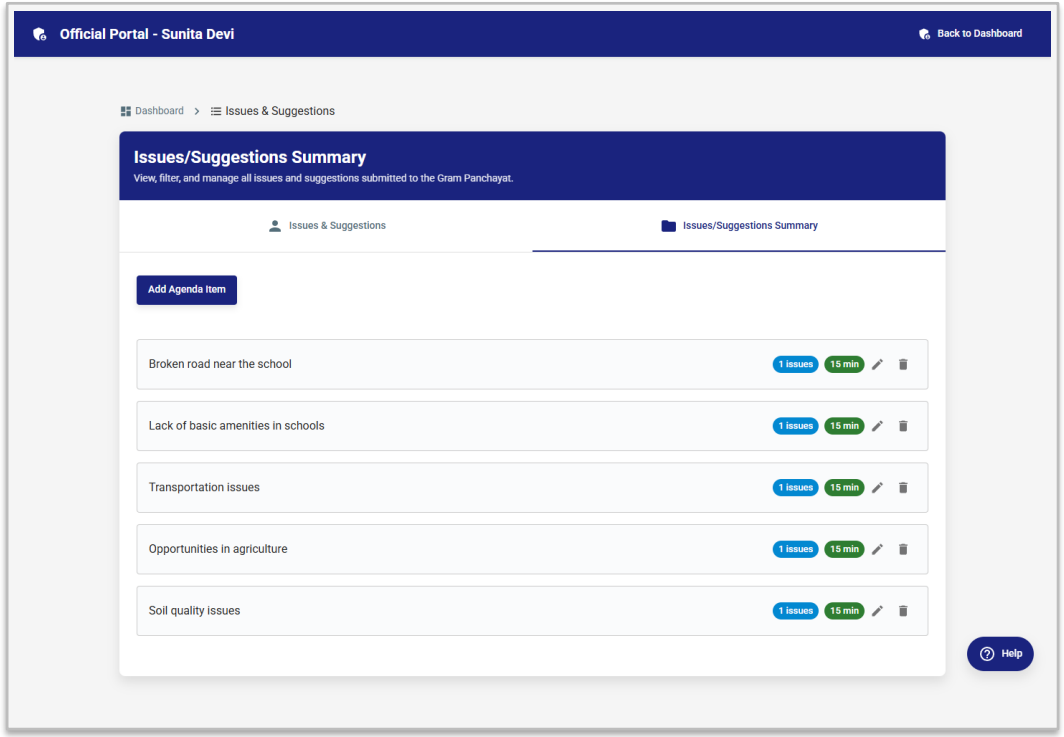
Official Portal

Issues / Suggestions Summary

AI-generated summaries of citizen issues, grouped by category and deduplicated. Bedrock Nova Lite runs hourly via cron to summarise new issues into agenda-ready items. Officials select items from this backlog for the next Gram Sabha agenda.

Key Highlights

- Hourly AI summarisation via Bedrock Nova Lite cron job
- Category-wise grouping with deduplication of similar issues
- One-click selection of items for Gram Sabha agenda



Official Portal — Issue Details

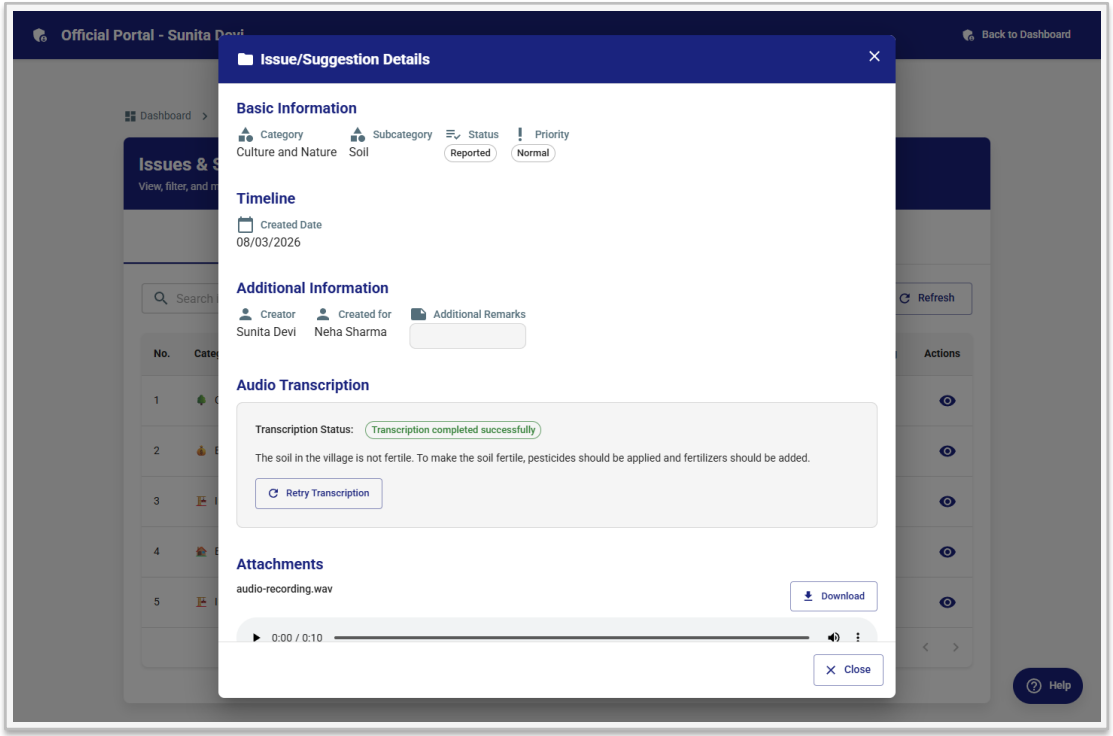
Official Portal

Issue Details Modal

Detailed view of a single issue showing the original voice transcription, AI-enhanced versions in 3 languages, sentiment analysis results, audio playback, and the full issue lifecycle history.

Key Highlights

- Original + enhanced transcription in English and Hindi
- Embedded audio player for original voice recording
- Sentiment scores, key phrases, and priority justification



Official Portal — Gram Sabha Management

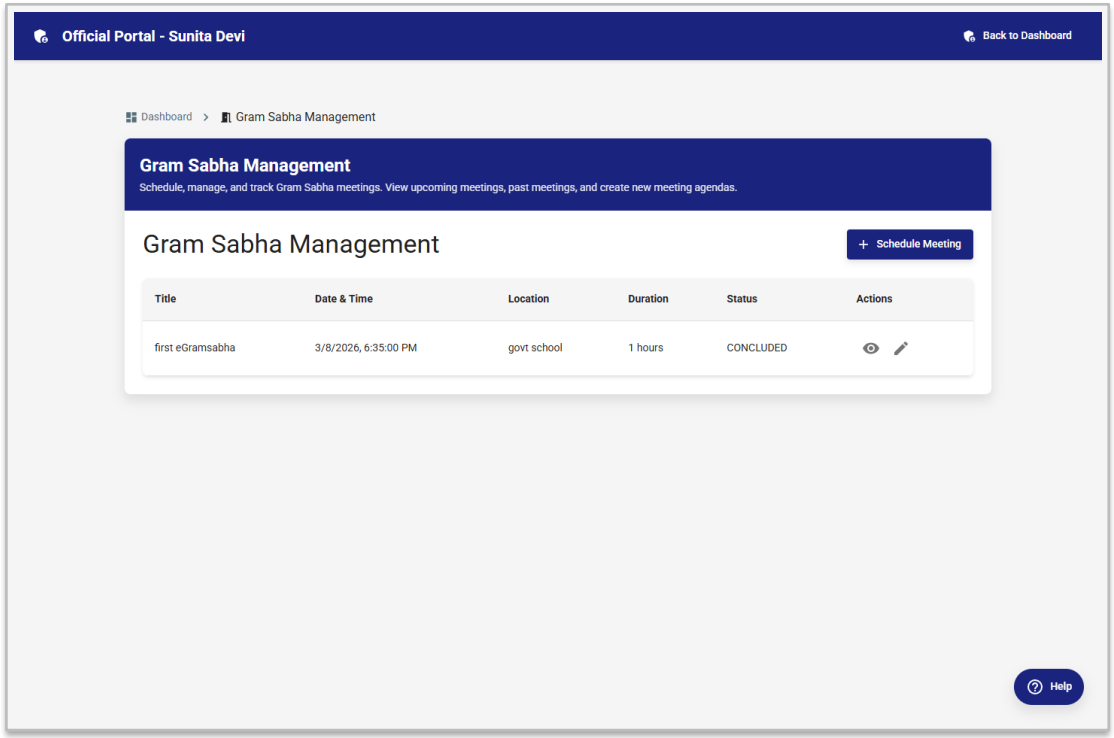
Official Portal

Gram Sabha Meetings

Lists all scheduled, in-progress, and completed Gram Sabhas. Officials can create new meetings, set quorum criteria, manage agendas, and view attendance reports from this central meeting management screen.

Key Highlights

- Meeting lifecycle: SCHEDULED → IN_PROGRESS → COMPLETED
- Quorum tracking with automatic status transitions
- PDF export for attendance reports and agenda documents



Official Portal — Gram Sabha Details

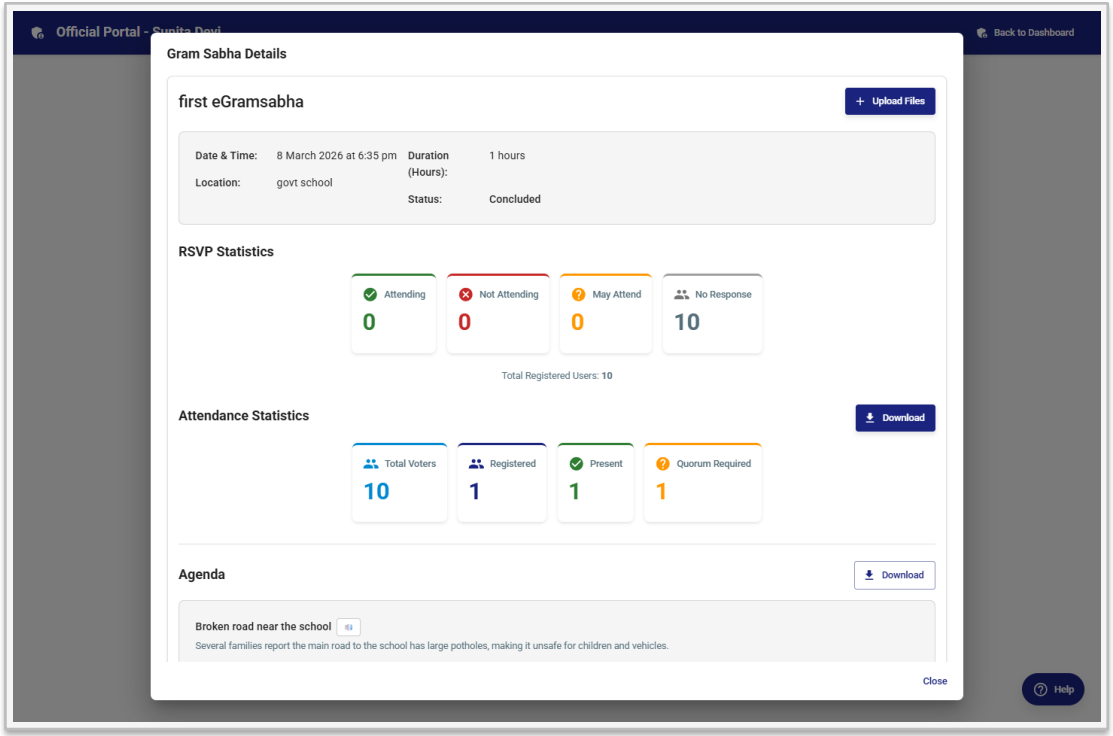
Official Portal

Gram Sabha Details

Detailed view of a specific Gram Sabha showing date, time, location, expected attendance, quorum status, assigned agenda items, and attendance records. Officials can manage the meeting lifecycle from this modal.

Key Highlights

- Meeting metadata: date, time, venue, expected duration
- Agenda items linked to citizen issues with AI summaries
- Attendance count vs quorum threshold indicator



Official Portal — Schedule Meeting

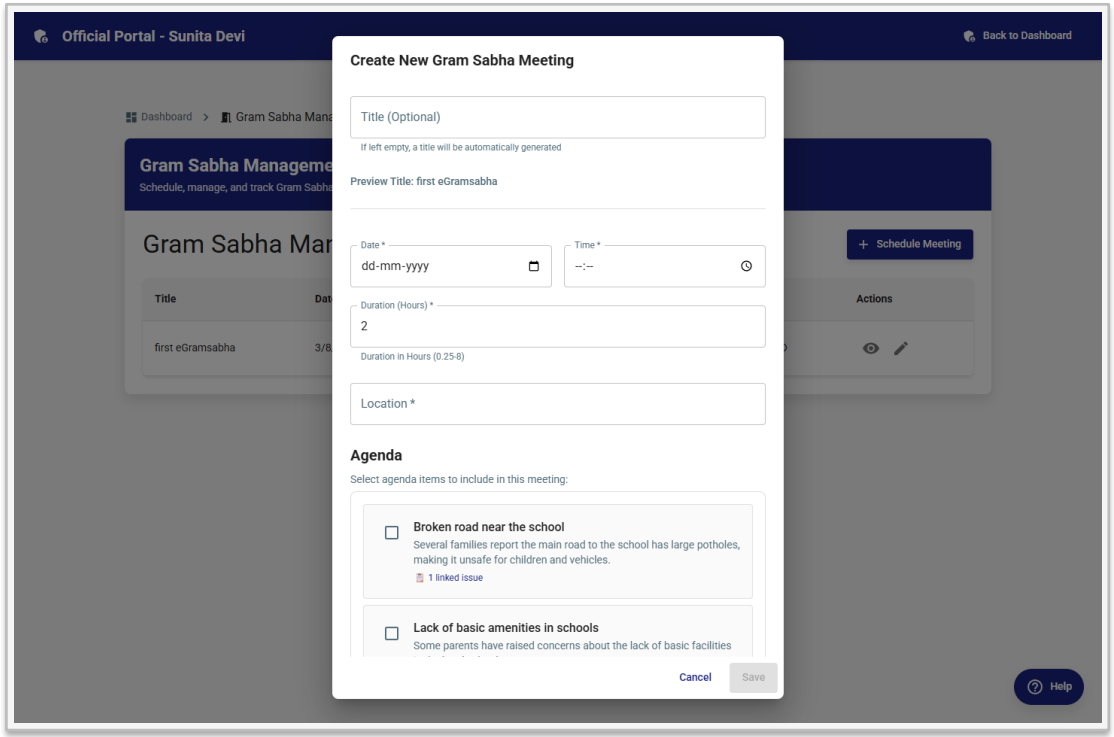
Official Portal

Create New Gram Sabha

Modal for scheduling a new Gram Sabha meeting. Officials set the date, time, location, and expected duration. SMS notifications are sent to registered citizens once the meeting is created.

Key Highlights

- Date, time, and venue selection for new meetings
- SMS notification to citizens via Amazon SNS (TRAI DLT ready)
- Auto-populates panchayat details from admin configuration



Official Portal — Biometric Attendance

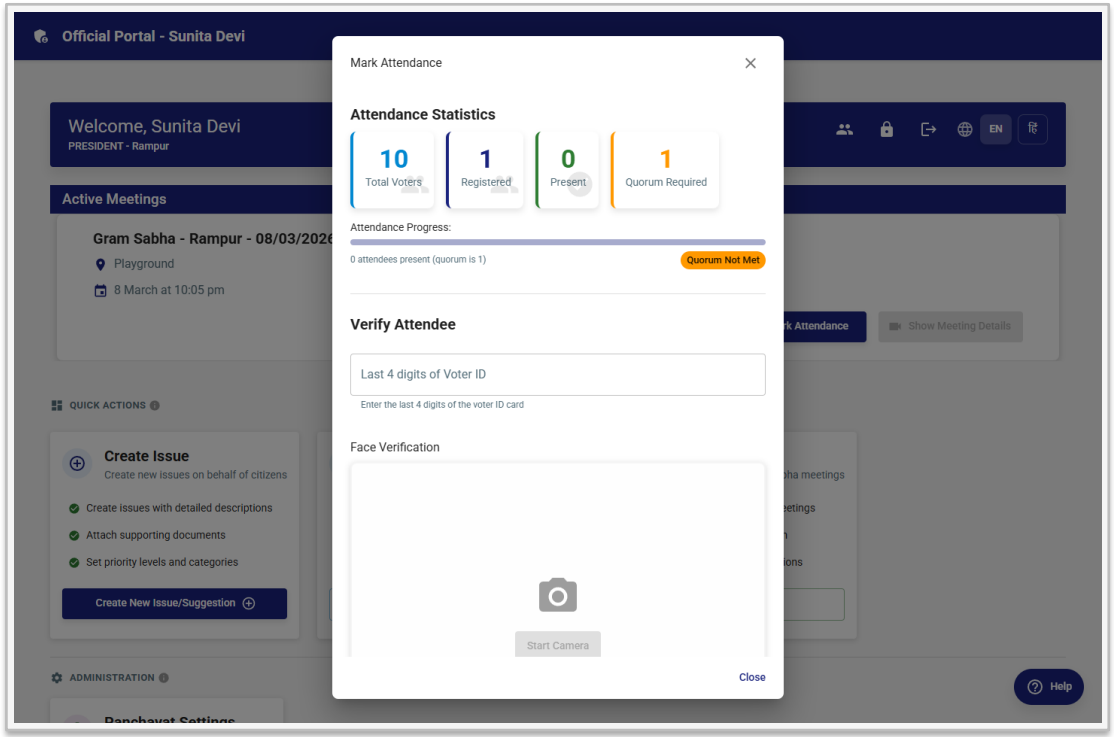
Official Portal

Mark Attendance

Face-verified attendance marking during Gram Sabha meetings. The camera captures the citizen's face and matches it against their registered 128-dim face descriptor. Verification method (face/manual) is recorded for audit.

Key Highlights

- Face verification with Euclidean distance < 0.4 threshold
- Quorum auto-check: meeting starts when threshold met
- Eliminates paper sign-in fraud completely



Official Portal — Attendance Confirmed

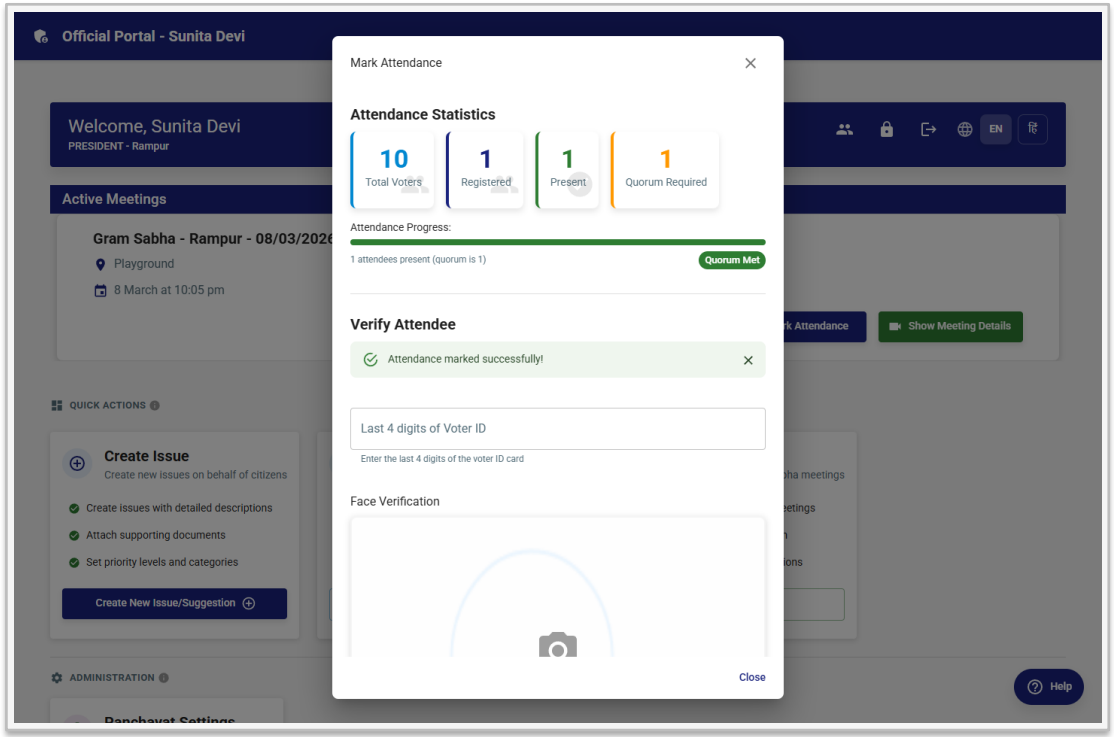
Official Portal

Attendance Marked Successfully

Success confirmation after marking attendance via face verification. Shows the matched citizen's name, verification method used, and updated attendance count vs quorum threshold.

Key Highlights

- Visual confirmation with citizen name and photo match
- Verification method recorded: face-verified or manual
- Live attendance count updates toward quorum threshold



Official Portal — Meeting Details

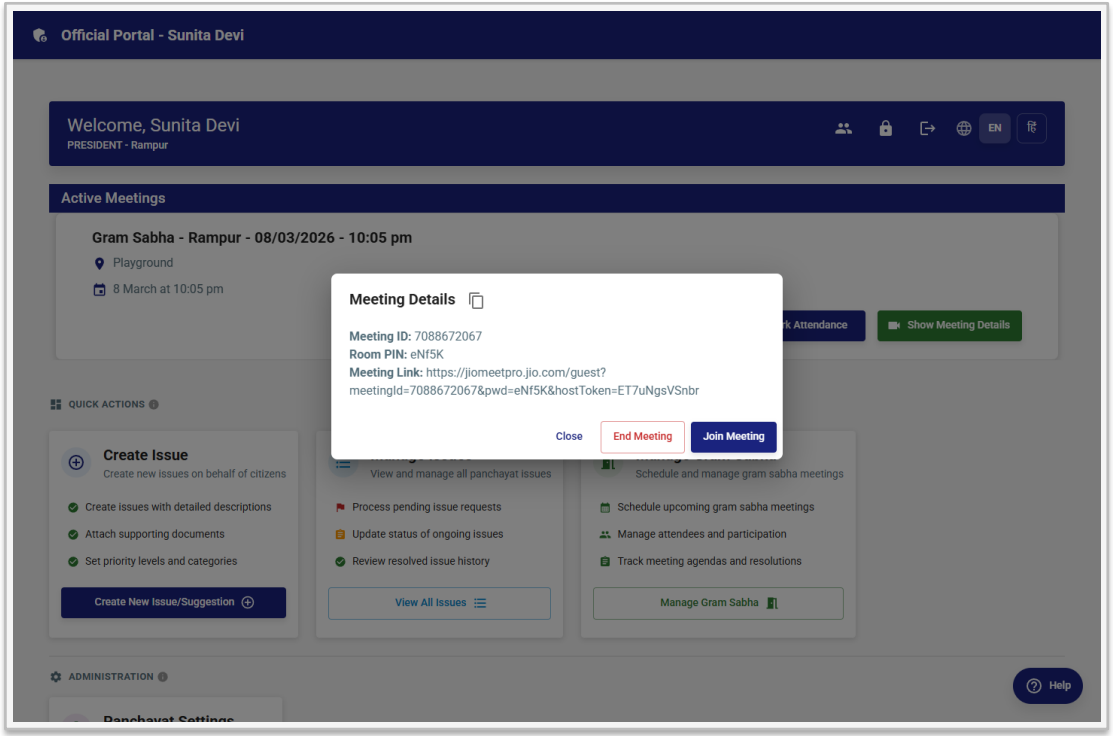
Official Portal

Meeting Details

Comprehensive meeting details modal showing the full meeting record: attendance list, agenda items discussed, decisions recorded, and resolution remarks. Officials can mark agenda items as discussed and add resolution notes.

Key Highlights

- Full attendance list with verification method per citizen
- Agenda items with discussion status and resolution remarks
- Post-meeting: undiscussed items return to backlog automatically



Official Portal — Agenda Selection

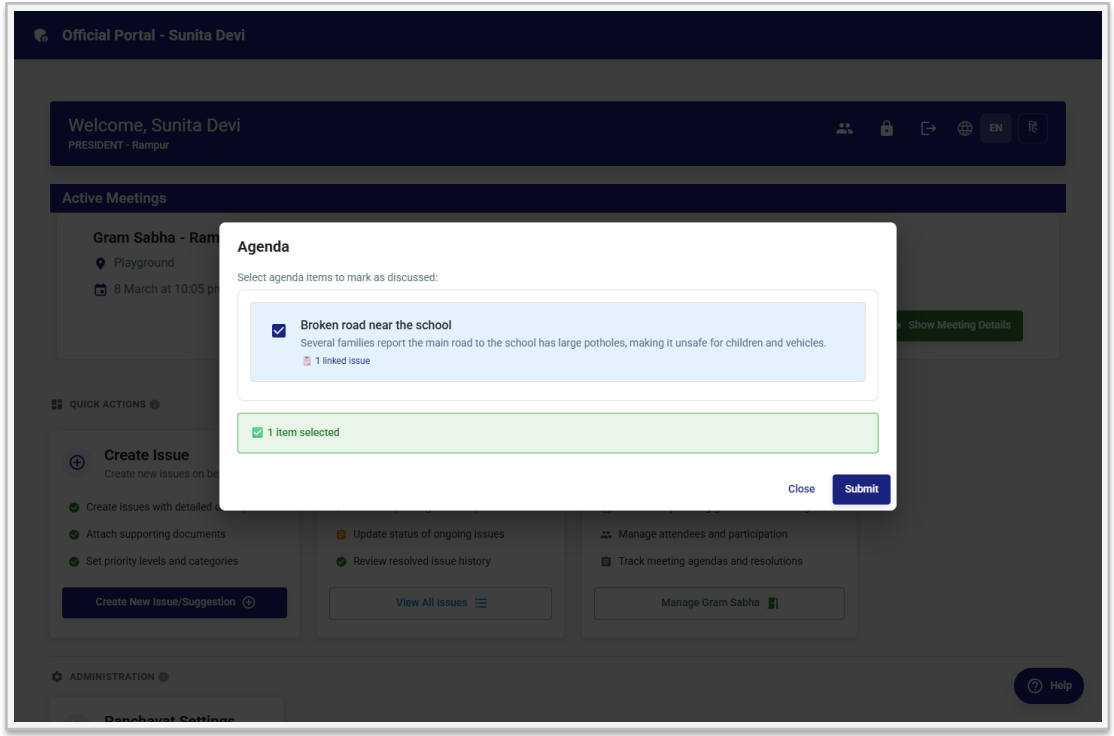
Official Portal

Agenda Item Selection

Officials select which AI-summarised issues to include in the Gram Sabha agenda. Selected items are structured into numbered agenda points with citizen context. The agenda can be translated and read aloud via TTS in multiple languages.

Key Highlights

- Select from AI-summarised issue backlog for agenda
- Mark agenda items as discussed during the meeting
- Translate agenda to Hindi + regional languages with TTS



Admin Portal — Login

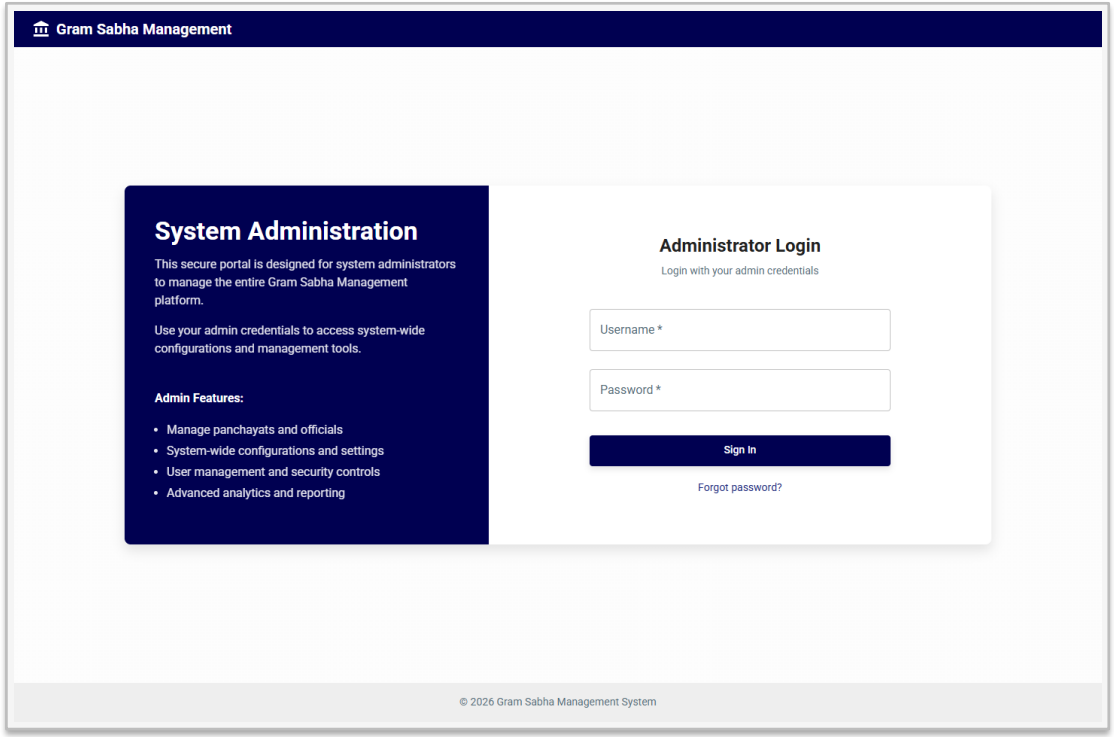
Admin Portal

Admin Login

The Admin Portal login page. Admins have the highest privilege level with access to platform configuration, panchayat CRUD, official management, and system-wide settings. JWT token with 12-hour expiry.

Key Highlights

- Highest privilege role with full platform access
- 12-hour JWT token with 7-day refresh token
- Access to all panchayats and system configuration

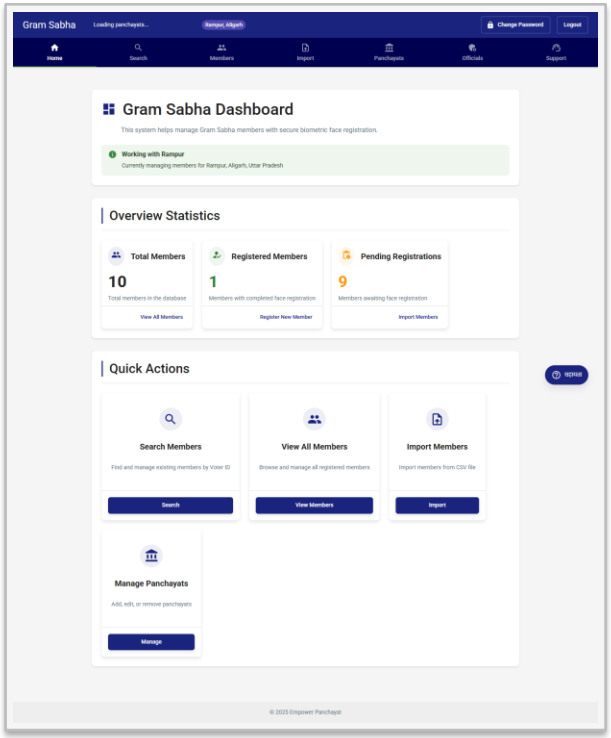


Admin Portal — Dashboard

Admin Portal

Admin Dashboard

The admin's central hub for platform management. Shows system-wide statistics, quick actions for panchayat creation, official management, and bulk CSV import. Admins manage the entire multi-panchayat hierarchy from here.



Key Highlights

- System-wide stats: total panchayats, officials, citizens
- Quick actions: Add Panchayat, Register Officials, Import CSV
- Support ticket management for citizen/official help requests

Admin Portal — Member Search

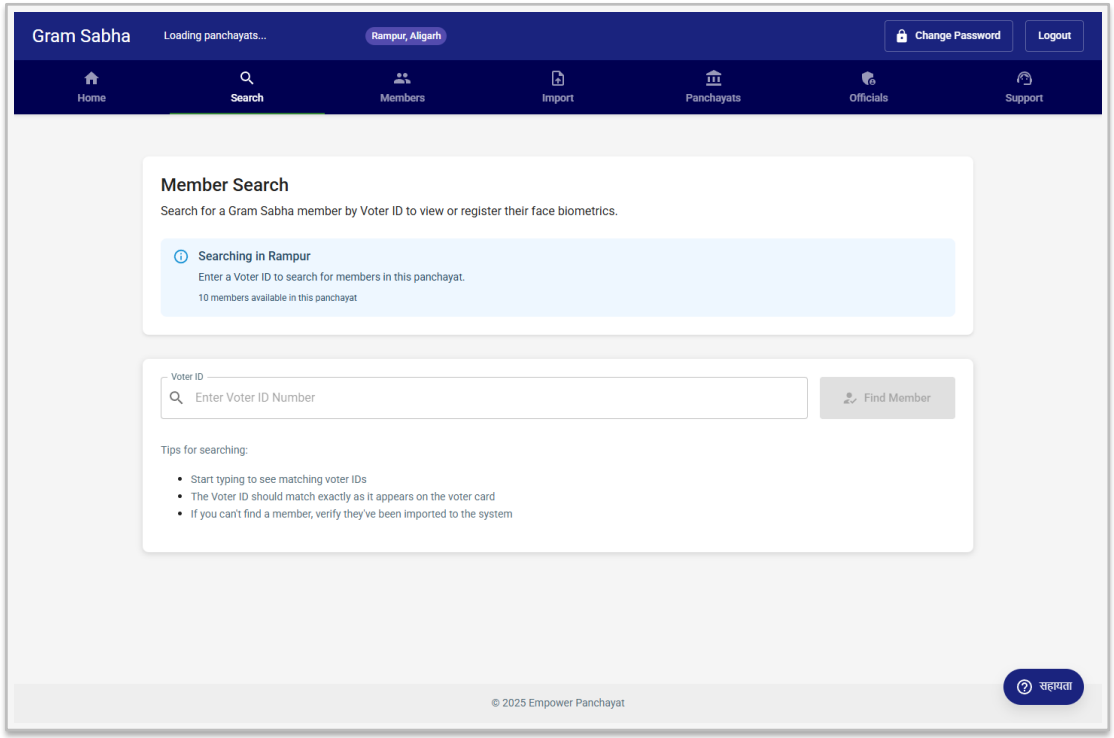
Admin Portal

Member Search

Search for registered members across all panchayats by name, voter ID, or phone number. Results show member details, registration status, and face registration status. Admins can click through to view or edit any member's profile.

Key Highlights

- Cross-panchayat search by name, voter ID, or phone
- Registration and face biometric status indicators
- Quick navigation to member profile for edits



Admin Portal — Member Profile

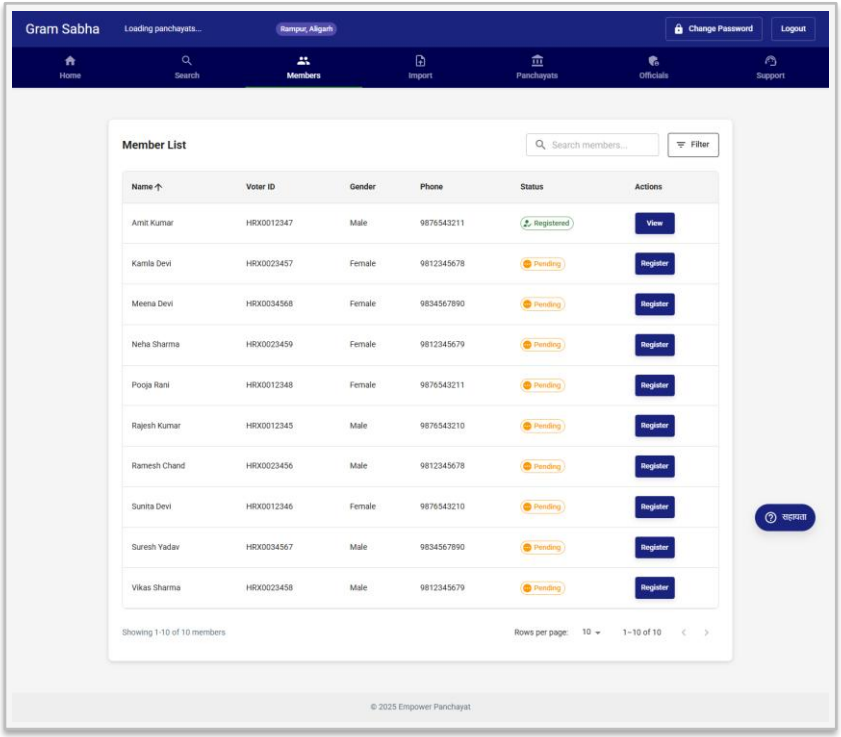
Admin Portal

Member Profile

Detailed member profile showing personal information, panchayat assignment, ward, and face registration status. Admins can capture or update the member's face biometric with liveness detection (2 blinks + 5 head movements).

Key Highlights

- Face registration with liveness detection via MediaPipe
- 128-dim face descriptor stored for future authentication
- Edit member details: name, voter ID, phone, ward assignment

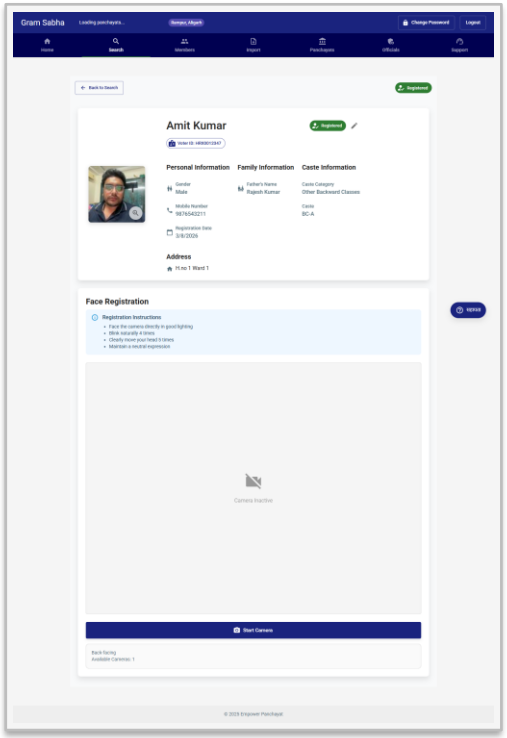


Admin Portal — All Members

Admin Portal

All Members

Complete list of all registered members across all panchayats. Sortable and filterable by panchayat, ward, and registration status. Shows face registration status and quick action buttons.



Key Highlights

- View all members with face registration status
- Filter by panchayat, ward, and registration status
- Pagination for large datasets with sort options

Admin Portal — CSV Import

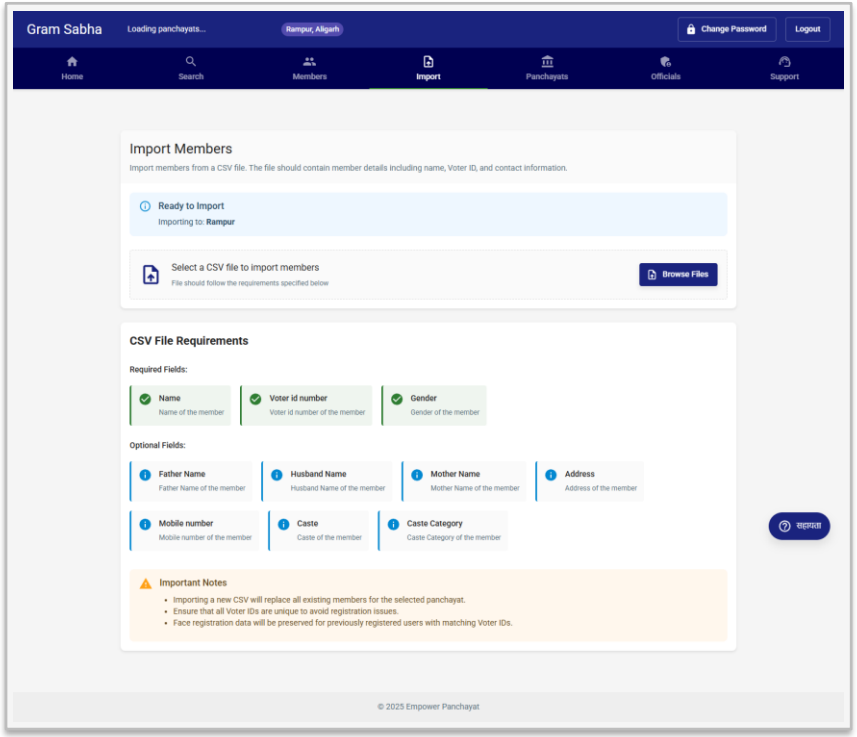
Admin Portal

CSV Import

Bulk import of citizens, officials, or panchayats via CSV upload. Shows required field specifications and validation rules. Batch processing with error reporting for failed rows.

Key Highlights

- Bulk import for citizens, officials, and panchayats
- Required field specifications and format validation
- Error reporting with row-level details for failed imports



Admin Portal — Panchayat Management

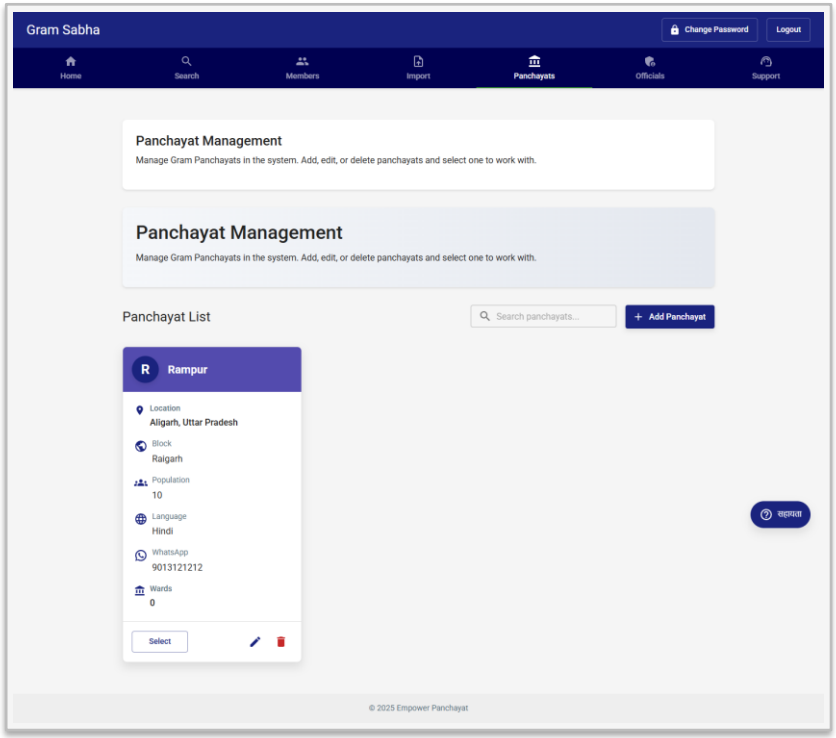
Admin Portal

Panchayat Management

Create, edit, and manage panchayats with LGD codes and ward structures. Each panchayat has its own letterhead, quorum criteria, and official assignments. Supports the full hierarchy from district to ward level.

Key Highlights

- CRUD operations with LGD code validation
- Ward management and letterhead upload
- Sabha quorum criteria configuration per panchayat



Admin Portal — Add Panchayat

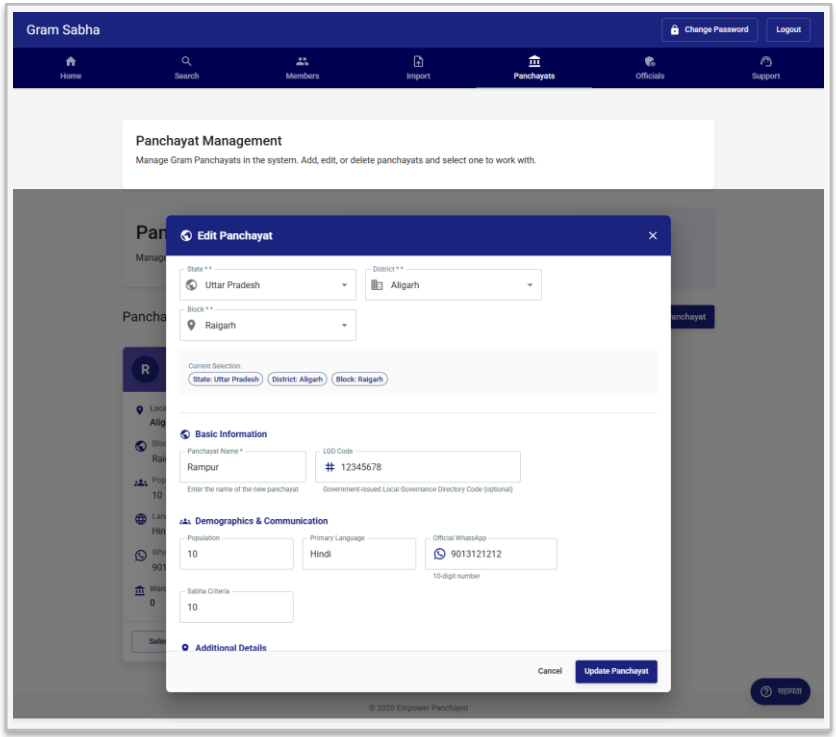
Admin Portal

Add New Panchayat

Modal for creating a new panchayat in the system. Captures essential details including panchayat name, LGD code, district, block, state, and contact information. Ward structure is configured after initial creation.

Key Highlights

- LGD code integration for official government mapping
- District, block, and state hierarchy selection
- Contact details and location information capture

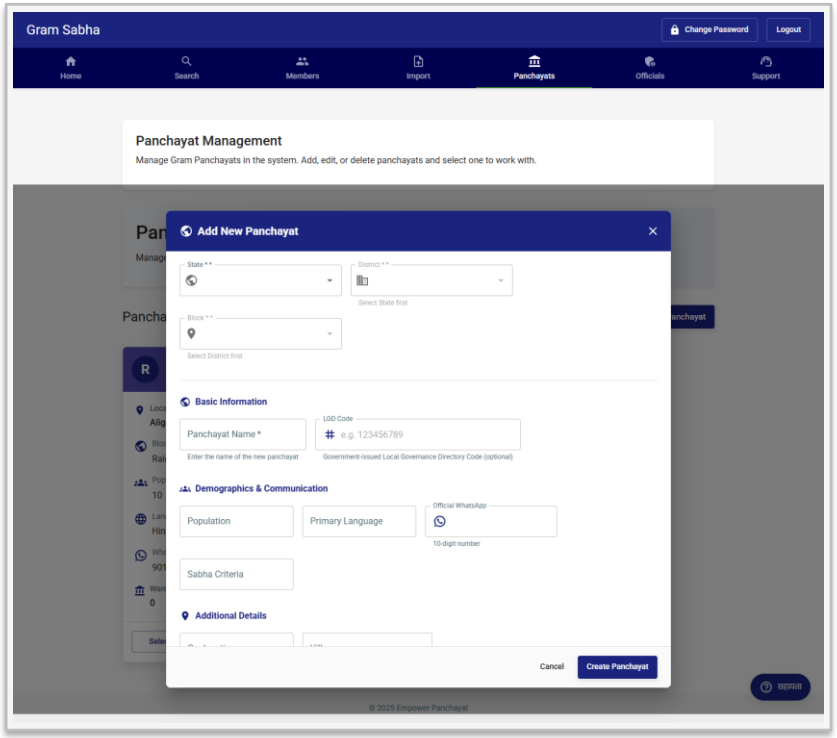


Admin Portal — Panchayat Form (Continued)

Admin Portal

Add Panchayat — Extended Fields

Continuation of the Add Panchayat form showing additional configuration fields including letterhead upload, quorum criteria percentage, ward count, and Sabha-specific settings.



Key Highlights

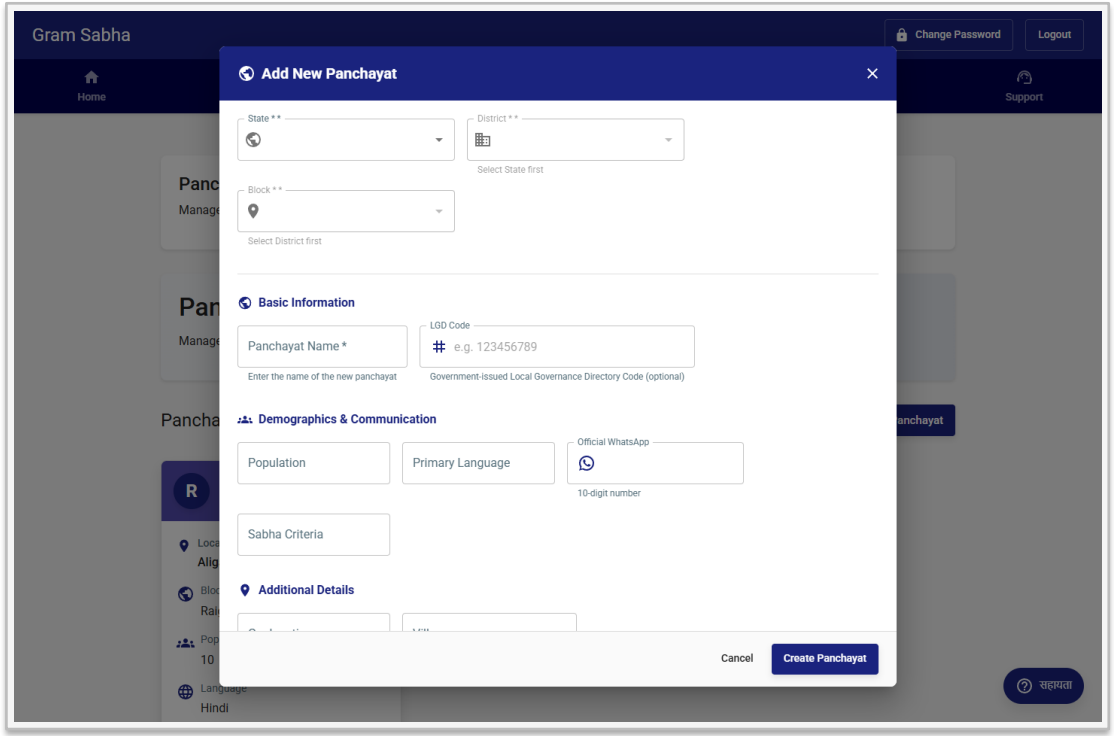
- Letterhead upload for branded PDF exports
- Quorum criteria percentage for Sabha validation
- Ward count and structure configuration

Admin Portal — Panchayat Form (Complete)

Admin Portal

Add Panchayat — Complete Form

The complete Add Panchayat form ready for submission, showing all fields populated. Includes validation indicators for required fields and a preview of the panchayat configuration before creation.



Key Highlights

- Form validation with required field indicators
- Preview of full panchayat configuration
- Submit to create panchayat and initialize ward structure

Admin Portal — Edit Panchayat

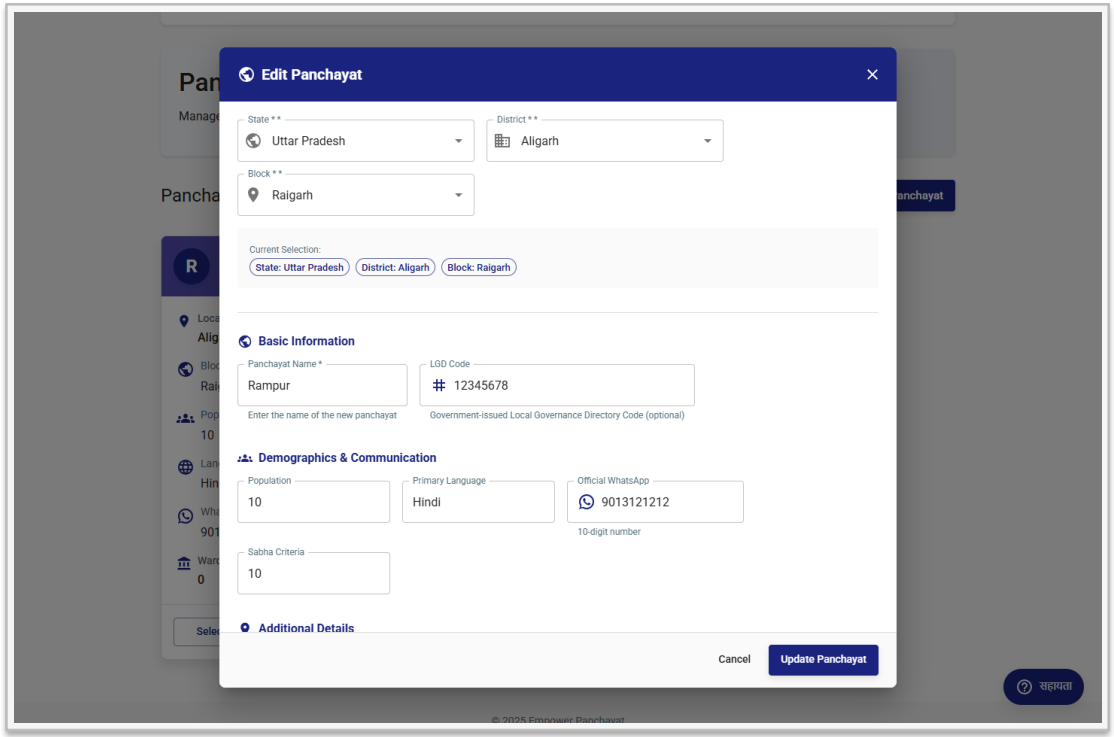
Admin Portal

Edit Panchayat

Edit dialog for updating an existing panchayat's details. All fields from creation are editable, including letterhead, quorum criteria, and contact information. Changes are applied immediately.

Key Highlights

- Edit all panchayat fields including LGD code and hierarchy
- Update letterhead and quorum criteria
- Immediate save with validation



Admin Portal — Official Management

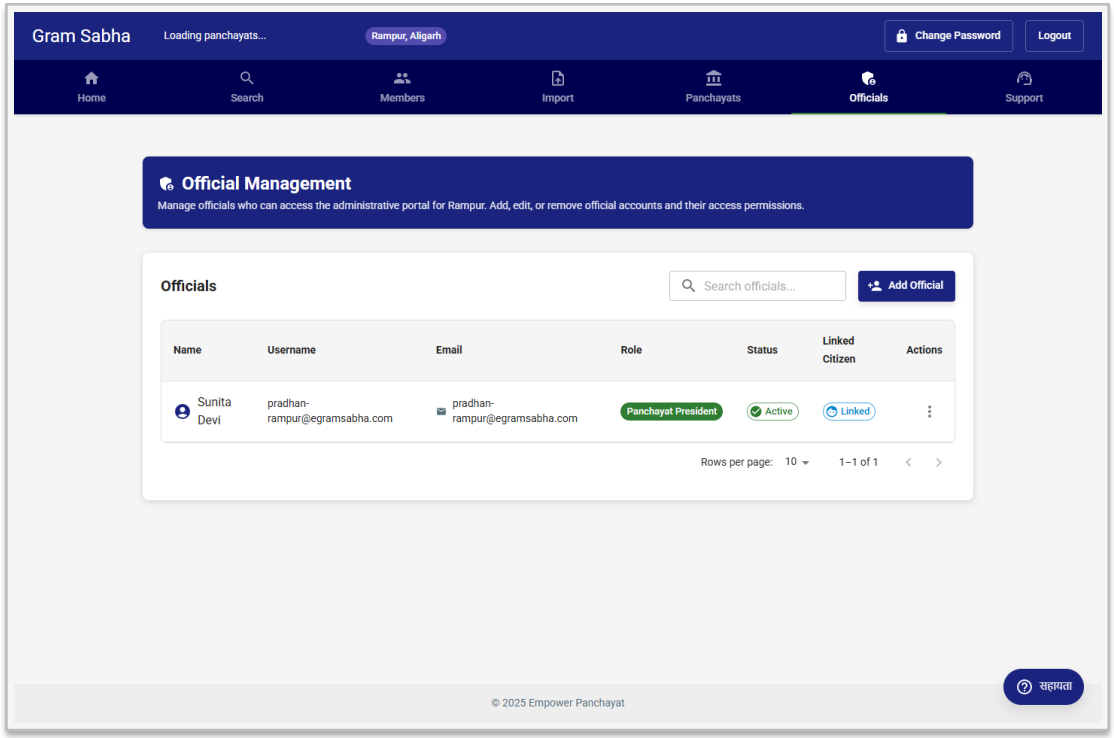
Admin Portal

Official Management

List of all registered officials across panchayats with their roles, assigned panchayat, and status. Admins can register new officials, edit roles, and manage access from this screen.

Key Highlights

- 6 official roles: Secretary, President, Ward Member, Committee Secretary, Guest, Admin
- Role assignment and panchayat linkage
- Edit or deactivate official accounts



Admin Portal — Edit Official

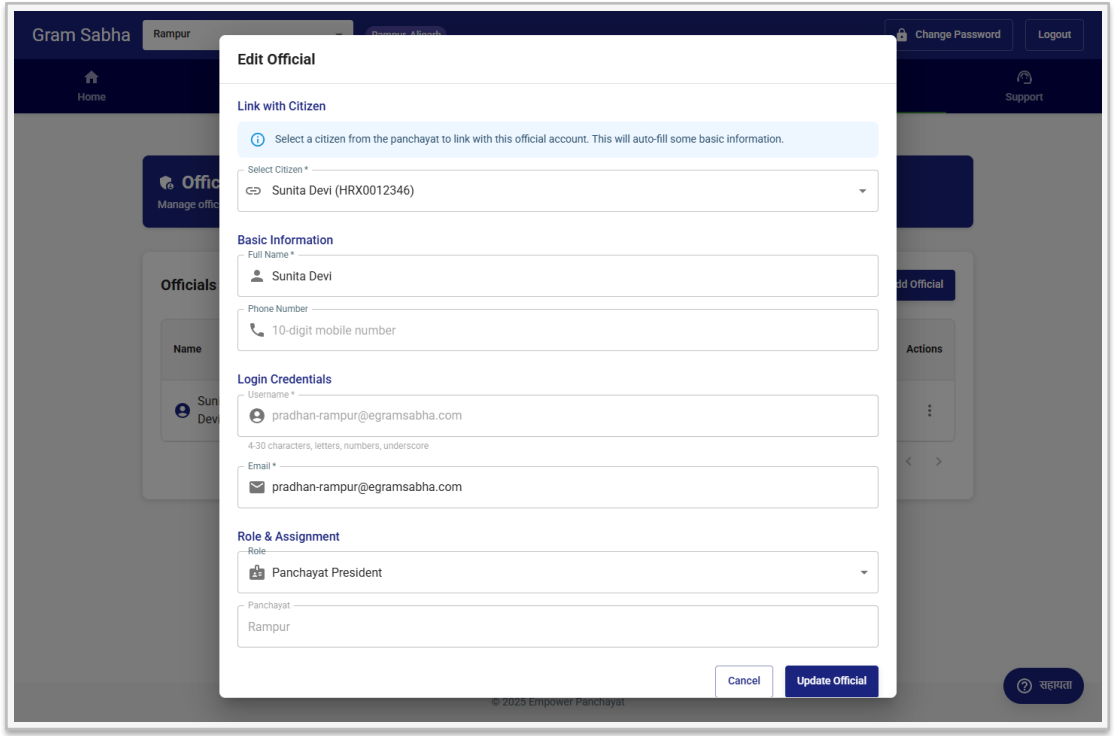
Admin Portal

Edit Official Form

Form for editing an official's details including name, role, assigned panchayat, contact information, and access level. Role changes take effect immediately and update the official's JWT permissions.

Key Highlights

- Role reassignment with immediate JWT permission update
- Panchayat and ward assignment management
- Contact details and credential management



Future Roadmap

Phase 1 Immediate

- Activate AWS Transcribe
- TRAI DLT for SNS SMS
- Enable Rekognition

Phase 2 Short-term

- DocumentDB migration
- MOM full integration
- eSign (Planned)
- Mobile app (React Native)
- e-Gram Swaraj integration

Phase 3 Medium-term

- Multi-panchayat escalation
- Analytics dashboards
- Video Gram Sabha
- Citizen self-attendance

Phase 4 National

- 250K panchayats rollout
- Aadhaar integration
- Open API for CSOs
- Scheme delivery tracking

eGramSabha — AI-Powered Digital Gram Sabha Platform

"Sashakt Panchayat, Samriddh Bharat"

Strong Panchayat, Prosperous India

GitHub: https://github.com/anshuljain90/ai4bharat-empowerpanchayat
Demo Video: https://drive.google.com/drive/folders/1aLeLtSPlDUaoypi3TAu2tg4I-TpxVUNH
Live Demo: https://44.194.253.143/
Admin Login: `admin / AdminPassword123`
Official Login: `password same as configured username`

Apache 2.0 | Digital Public Good | Open Source

Innovation partner H2S HACK2SKILL | Media partner YOURSTORY

AI for Bharat Hackathon

Powered by aws

Thank You